

Stable and recurrent deformation subspaces preserve pelvic mechanical function across anatomical variability

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Short title: Recurrent pelvic deformation subspaces

Abstract

Human pelvic kinematics must operate across marked variation in ring geometry, inlet and outlet dimensions, and bone mineral distribution. Yet, population computational biomechanics commonly compares ranked deformation pathways as if identical spectral labels carried identical anatomical meaning across individuals. Here, we investigate whether pelvic deformation patterns remain attached to individually stable pathways or are preserved within compact recurrent deformation subspaces. We analyse the linearised stiffness spectrum of $N=278$ subject-specific finite-element models assembled on a shared reference domain with individual bony anatomy and CT-derived elasticity. Across the population cohort, pelvic deformation partitions into two distinct spectral regimes. Modes 1–3 form a gap-protected low-rank *backbone* that rarely reorders across subjects ($< 0.4\%$ label-exchange rate) and governs habitual standing loads. Higher ranks contain recurrent *clustered* regimes, most prominently an inlet-related 9–10 block present in 54.7% of individuals. Within that block, anterior–posterior inlet coupling redistributes between rank labels 9 and 10 when the labels exchange, yet remains consistently high in their shared two-dimensional subspace and is selectively recruited under parturition-motivated proxy loading. An outlet-related 5–6 block similarly forms an outlet-related subspace with strong biischadic coupling despite local label instability. These findings demonstrate that pelvic compliance is organised across spectral scales: low-rank pathways provide a stable mechanical backbone, whereas higher-order anatomical deformation programs are retained within compact recurrent modal subspaces. More generally, this framework establishes how anatomical coupling can be preserved across skeletal variability without assuming one-to-one conservation of individual spectral labels.

Author summary

Individuals differ substantially in pelvic shape and bone density, yet every human pelvis must transfer body weight during locomotion while preserving internal canal dimensions. Biomechanical studies typically compare individuals by examining vibration or deformation modes one by one, assuming that each ranked pattern corresponds to the same mechanical behavior in every person. Using detailed computational finite-element models of 278 human pelvises, we tested whether this assumption holds across population anatomy. We found that everyday standing loads rely on a few dominant, stable deformation patterns (modes 1–3) that maintain consistent rank ordering and strong shape alignment across individuals (median mass-weighted modal assurance criterion to reference > 0.96). In contrast, patterns associated with widening of the pelvic inlet do not remain attached to a single fixed numerical rank. Instead, this anatomical opening capacity remains remarkably stable across recurrent deformation subspaces whose internal mode order can swap between individuals. Comparing skeletons strictly mode-by-mode can therefore create the false impression of functional disruption when the underlying anatomical capacity is substantially more stable at the subspace level. Our study provides a principled computational approach to compare anatomically variable structures at the level of small groups of cooperating deformation pathways.

1 Introduction

Biological load-bearing structures must maintain mechanical competence across individuals that differ substantially in skeletal morphology and tissue distribution. The human pelvis exemplifies this challenge: it transfers locomotor forces between the spine and lower limbs while preserving birth canal dimensions critical for parturition [1–3]. A central computational question is how anatomical deformation pathways remain functionally coherent despite wide population variability.

In biological systems, robustness is achieved through the preservation of functional output under perturbation, frequently supported by degeneracy—the capacity of structurally distinct elements to perform overlapping tasks [4–7]. In structural biomechanics, transmitting body weight or accommodating internal cavity expansion need not rely on an identical, rigid deformation mode in every individual. Different pelves may achieve equivalent compliance through slightly different deformation pathways within shared subspaces, analogous to motor synergy recruitment across variable musculoskeletal anatomy [8].

Population computational biomechanics, however, frequently compares modal pathways across subjects under the assumption of stable spectral indexing. This convention conflates three distinct physical objects: (i) *anatomical shape*, describing subject-specific geometry; (ii) *deformation pathways*, describing physical ring kinematics; and (iii) *spectral rank*, which sorts pathways purely by ascending stiffness. In linearised elasticity, rank ordering is robust only while adjacent eigenvalues remain well separated. When eigenvalues crowd, minor perturbations in geometry or tissue density rotate eigenvectors and induce label exchange (veering) even while the physically relevant subspace remains unchanged [9–11]. Assuming one-to-one correspondence between mode rank and biological function across a cohort therefore introduces a severe risk of misinterpreting local label swaps as genuine functional disruption.

The appropriate unit of analysis in crowded spectral regimes is not the isolated eigenmode, but a *compact recurrent subspace*—a low-dimensional, externally separated block that retains a targeted deformation program across subjects. The pelvic ring offers an ideal anatomical testbed: its compliance is constrained by joint morphology, articular cartilage, and extensive ligamentous networks [12–14], yet its population-scale modal stability has remained unexplored.

Here, we evaluate the linearised stiffness spectrum of $N=278$ CT-derived, subject-specific finite-element models on a common reference domain to address three testable computational questions:

- (i) **Rank stability:** Are primary pelvic deformation pathways order-locked across a variable population cohort, or do they reorder systematically?
- (ii) **Subspace preservation:** In crowded spectral regimes, do anatomical compliance observables redistribute between mode labels while remaining stable within compact recurrent subspaces?
- (iii) **Load-routing selectivity:** Do habitual standing and parturition-motivated proxy loads recruit distinct modal regimes, separating conserved baseline support from flexible higher-order pathways?

2 Results

2.1 Interpretable deformation pathways preview two distinct organisational scales

Before turning to cohort statistics, it was investigated whether the low-order deformation repertoire already reveals where mechanical meaning is stably localised and where it is likely to be preserved only at the level of a small clustered family. Figure 1 characterises the global deformation of the complete reference pelvis without separating bones or anatomical branches. Direct fitting of simple kinematic templates to the local strain field leaves 97–99% of the norm unexplained across all modes, because compliance and strain heavily concentrate in the cartilaginous joints (sacroiliac joints and pubic symphysis), which drowns out global ring deformation and fails to discriminate modes. To recover the global ring-level character without imposing artificial 1D beam axes on a closed 3D pelvic ring, we evaluate the overall shape change across the entire 3D continuum: infinitesimal rigid-body translation and rotation are first mathematically removed by orthogonal projection, after which the remaining nonrigid displacement field

is decomposed into four canonical 3D deformation families (axial tension/compression, bending, transverse shear, and torsion). Rigid rotation is eliminated upfront and is not a reported deformation category. Subspace overlap on the irregular geometry is handled via Shapley allocation of the explained squared nonrigid shape norm; unexplained deformation is retained as an explicit residual. The resulting fractions are descriptive shape fractions conditional on the kinematic dictionary, not strain-energy shares. The complete mathematical formulation, positive 14-point (degree-4) tetrahedral quadrature, 24-dimensional polynomial dictionary, Gram matrix conditioning, synthetic validation benchmarks, and complete 15-mode quantitative profile tables are detailed in Appendix E (Tables E1–E2). Population stability is tested separately below.

Higher modes raise a different biological question: whether a mechanically interpretable function remains tied to one label or only to a compact family. The most important case is the 9–10 pair. Mode 9 typically exhibits an inlet-opening deformation in which the pubic symphysis and the sacral promontory move apart, widening the pelvic ring. Mode 10 is a near-degenerate partner with an asymmetric field concentrated around the pubic region. Their close spacing already suggests that inlet-related behaviour may not be consistently attached to one fixed rank label across subjects, even if it remains remarkably stable within the two-dimensional span they share. For this region of the spectrum, the pair is therefore a more reliable biological interpretive unit than either label alone.

Whole-pelvis 3D canonical deformation patterns



Figure 1: **Whole-pelvis deformation patterns of reference modes.** Anterior (left) and cranial (right) views show displacement magnitude from purple to yellow, scaled to a 50 mm maximum; grey geometry is undeformed. Each bar partitions the whole-pelvis nonrigid squared shape norm into fitted axial tension/compression, bending, transverse shear, torsion, and residual. Infinitesimal rigid-body translation and rotation are projected out upfront; rigid rotation is not a reported deformation category. The decomposition operates in full 3D Cartesian space across the entire pelvic ring without imposing a 1D reference axis. Overlapping pattern subspaces are allocated using Shapley values. “Fitted” reports the percentage of nonrigid shape change explained by the four canonical 3D families. All fractions use the complete continuum without bone-wise subdivision and describe kinematic shape change rather than elastic-energy shares.

2.2 Population variability separates an order-locked backbone from recurrent clustered regimes

The cohort of $N=278$ subjects resolves the spectrum into two regimes: an order-locked low-rank backbone and recurrent higher-rank clustered blocks (Fig. 2, Table 1). Modes 1–3 are rarely reordered across the cohort (swap rate $< 0.4\%$), whereas higher ranks contain small consecutive gaps and frequent label exchange. Using a data-driven cluster threshold $\varepsilon_{\text{in}} \approx 0.131$ (25th percentile of pooled relative consecutive gaps; see Methods), we detect nine near-degenerate rank clusters above a 3% prevalence floor. The most prevalent cluster is rank modes 9–10 (54.7% of subjects), followed by rank modes 12–15 (41.7%) and rank modes 5–6 (33.8%). No eigenvalue pair met the operational numerical multiplicity criterion ($\text{gap}_{\text{in}} \leq 10^{-6}$). No obvious spatial symmetry was identified that would enforce repeated eigenvalues, although accidental degeneracy is not mathematically excluded.

Anatomical variability does not destabilise the spectrum uniformly. Label fragility is concentrated in a small number of compact families, while low-rank order is preserved. For the dominant 9–10 and 12–15 clusters, median relative internal gaps are small enough that modest perturbations can reverse local ordering. The 9–10 block is externally separated from both neighbouring modes, whereas the 12–15 regime is left-separated but right-truncated by the eigensolve; in both cases, the dynamics are best interpreted as compact clustered subspaces rather than as a global loss of spectral organisation. This pattern is consistent with a pelvic system that preserves stable baseline load-transfer pathways while allowing selected higher-rank functions to vary within a limited local repertoire.

Permutation statistics reinforce this interpretation. Modes 12 and 13 exhibit the highest swap rates, and mode 15 also shows a sizeable unpaired fraction under strict cross-subject pairing, whereas the backbone is essentially fixed in its ordering. For population computational biology the consequence is concrete: while one-to-one modal comparison is well-conditioned for the low-rank backbone, higher-rank comparisons must accommodate mode mixing and label exchange through subspace representations.

A variability decomposition into shape-only and material-only datasets (Fig. 2C) shows that shape variability accounts for the largest share of the coefficient of variation for modes 1–8, whereas bone-density heterogeneity contributes with increasing relative weight in modes 9–15. Shape changes alter *how* the spatial density heterogeneity enters the stiffness operator: shape dominates lower modal variance while material heterogeneity contributes increasingly at higher ranks. This structural reweighting preserves a conserved low-rank backbone while pushing selected higher-rank directions toward clustered, near-degenerate behaviour.

These prevalence estimates are not driven by a few outliers or by the order in which subjects are processed. Subsampling stability analysis (500 random draws of 80% cohort size without replacement) yields narrow 95% stability intervals for the dominant clusters (9–10: 51.8–57.7%; 12–15: 38.7–45.0%), confirming that cluster detection is robust to cohort composition. In addition, full subject resampling with replacement ($N=278$, 500 replicates) with threshold re-estimation confirms consistent prevalence estimates. Random permutation of subject order leaves cluster prevalence and swap statistics unchanged to machine precision. The central spectral hierarchy is therefore a cohort-level feature rather than a post-processing artefact. Additional cluster-mapping, swap, and robustness diagnostics are provided in Appendix A and Appendix B.

2.3 Clustered subspaces preserve distinct mechanical programs under variability

Once individual labels become unstable, the next question is whether the corresponding small recurrent subspaces remain mechanically distinct. Figure 3 shows that several do, when interpreted using invariant-subspace descriptors rather than individual eigenvector labels. Two

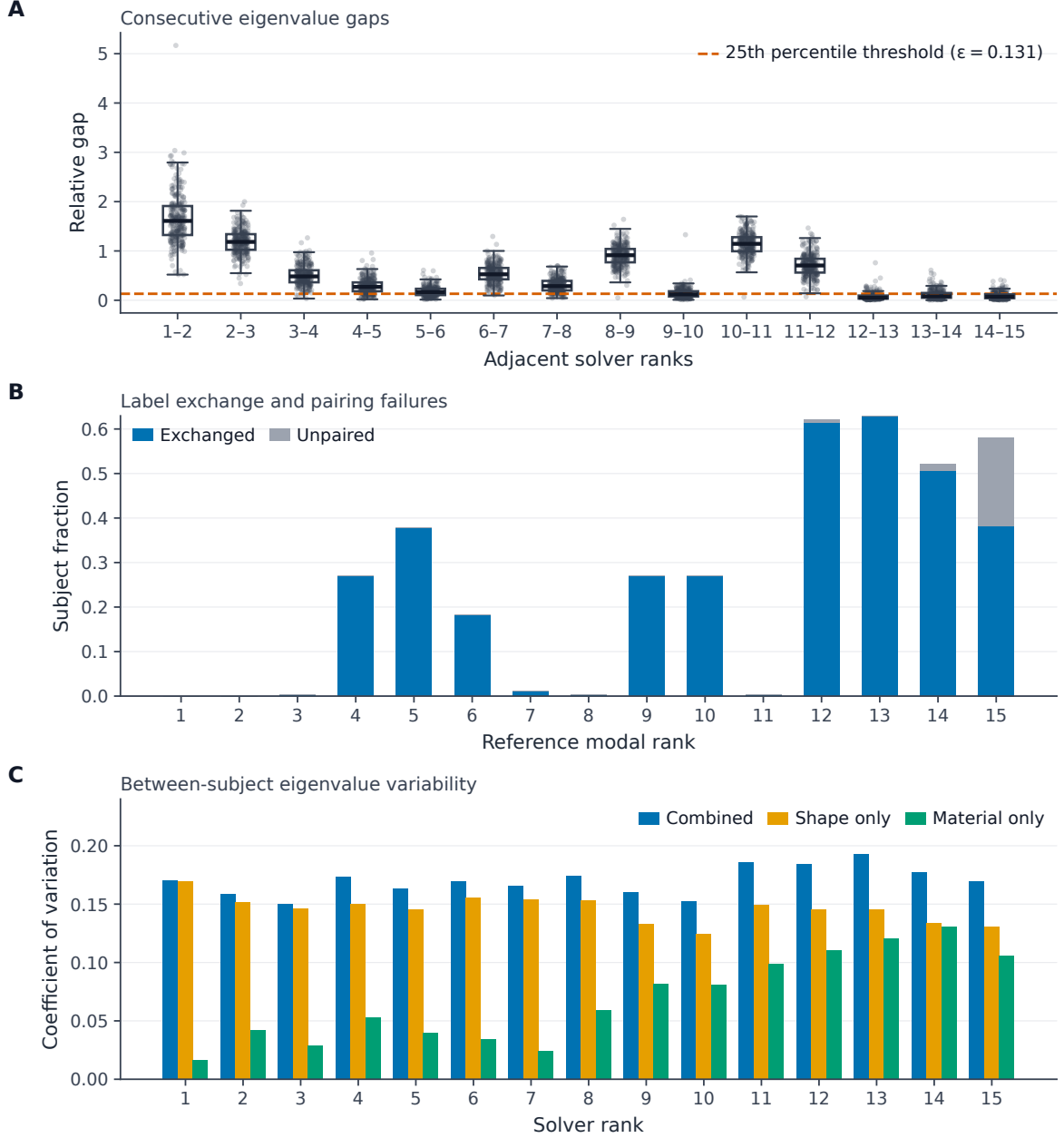


Figure 2: **Population variability partitions the stiffness spectrum into an order-locked backbone and clustered higher ranks.** (A) Consecutive relative spectral gaps $\text{gap}_{\text{in}}(i) = (\lambda_{i+1} - \lambda_i)/\lambda_i$ across the cohort ($N=278$); the dashed line marks the data-driven cluster threshold $\epsilon_{\text{in}} \approx 0.131$ (25th percentile). (B) Subject fraction exhibiting label exchange (blue) and unpaired pairing failures (grey; $\text{MAC}_{M_0} < 0.05$) per reference label. (C) Eigenvalue variability (coefficient of variation, CoV) across the combined cohort (blue), shape-only cohort (orange), and material-only cohort (green).

basis-invariant outputs were evaluated. First, Grassmann distance from a reference subject quantifies how much a subspace rotates across the cohort. Second, basis-invariant anatomical capacities for anatomical diameter functionals (AP and ML inlet; biischadic and bituberous outlet) provide functionally interpretable coordinates inside each clustered block (Methods).

The rank 9–10 subspace shows the strongest inlet-related capacity among the prevalent clusters. Its median AP-inlet capacity is 0.54 mm of inlet diameter change per 1 mm of maximum nodal displacement, making 9–10 the clearest higher-rank inlet-related program in the cohort.

Table 1: **Recurrent clustered rank blocks detected above the 3% prevalence floor.**

Cluster	Dim	n	Prev.	Med. gap _{in,min}	Med. sep _{out}	Type
Rank modes 9–10	2	152	54.7%	0.08114	0.4926	near-degenerate
Rank modes 12–15	4	116	41.7%	0.03153	NaN	near-degenerate
Rank modes 5–6	2	94	33.8%	0.08737	0.2345	near-degenerate
Rank modes 12–13	2	78	28.1%	0.03839	0.1941	near-degenerate
Rank modes 14–15	2	70	25.2%	0.05116	NaN	near-degenerate
Rank modes 12–14	3	45	16.2%	0.04109	0.1855	near-degenerate
Rank modes 4–5	2	35	12.6%	0.09521	0.2185	near-degenerate
Rank modes 13–15	3	29	10.4%	0.04465	NaN	near-degenerate
Rank modes 7–8	2	21	7.6%	0.1051	0.3676	near-degenerate

In contrast, the rank 5–6 cluster shows the strongest transverse outlet-related capacity, with median biischiadic widening 0.95 mm/mm and bituberous response 1.48 mm/mm (accompanied by 0.50 mm/mm for outlet AP dilation; in the 9–10 subspace, biischiadic widening reaches 1.20 mm/mm and outlet AP reaches 1.28 mm/mm). Different clustered subspaces therefore carry different mechanical programs: one is functionally aligned with the inlet, another with the outlet.

The four-dimensional 12–15 block illustrates a third regime. It has high prevalence, small internal gaps, and a richer pattern of within-block mixing than a simple two-mode exchange. Its anatomical capacity structure is more diffuse than in 9–10 or 5–6, consistent with it representing a broader, higher-order reserve rather than a single sharply interpretable program. Furthermore, because spectral eigensolves were truncated at $N_{\text{ev}} = 15$, external separation of this upper block on the right cannot be determined without higher eigenvalues; we therefore treat 12–15 as a diffuse higher-rank regime bounded by modal truncation. Not every recurrent cluster corresponds to one clean anatomical function. This heterogeneity is itself biologically informative: the framework separates compact, functionally specific blocks from mixed higher-rank families, and the former are the clearest targets for single-function biological interpretation.

Subspace orientation in the ambient displacement space (quantified by the Grassmann distance d_G , which is strictly invariant to internal basis rotations) also varies systematically with sex. The largest contrast is observed for the 9–10 block, indicating that female and male anatomies orient this two-dimensional inlet subspace along slightly different directions in $\mathbb{R}^n$. This qualifies how clustered behaviour is distributed across the population without altering the central point: recurrent clustered subspaces retain distinct mechanical meanings across sexes despite modest differences in their ambient subspace orientation.

2.4 Subspace-level capacity is substantially more stable than fixed labels in clustered regimes

A direct consequence of this framing follows: in a clustered regime, functional anatomical capacity is substantially more stable at the scale of the subspace than coupling attached to individual spectral ranks. This hypothesis was tested for the most prevalent cluster, rank modes 9–10 (Fig. 4).

Among subjects within the 9–10 cluster ($n=152$), AP-inlet coupling attached to *rank mode 9* is high when labels do not swap ($n=95$) and low when they do ($n=57$) (median 0.454 vs. 0.139 mm/mm; difference -0.315 mm/mm, 95% CI [-0.360, -0.262], $p = 2.7 \times 10^{-23}$, rank-biserial $r_{\text{rb}} = -0.97$). The complementary pattern appears for *rank mode 10* (median 0.163 vs. 0.436 mm/mm; difference +0.273 mm/mm, 95% CI [0.225, 0.335], $p = 1.7 \times 10^{-22}$, $r_{\text{rb}} = 0.95$). Label-level coupling therefore redistributes decisively between the two rank indices, not because

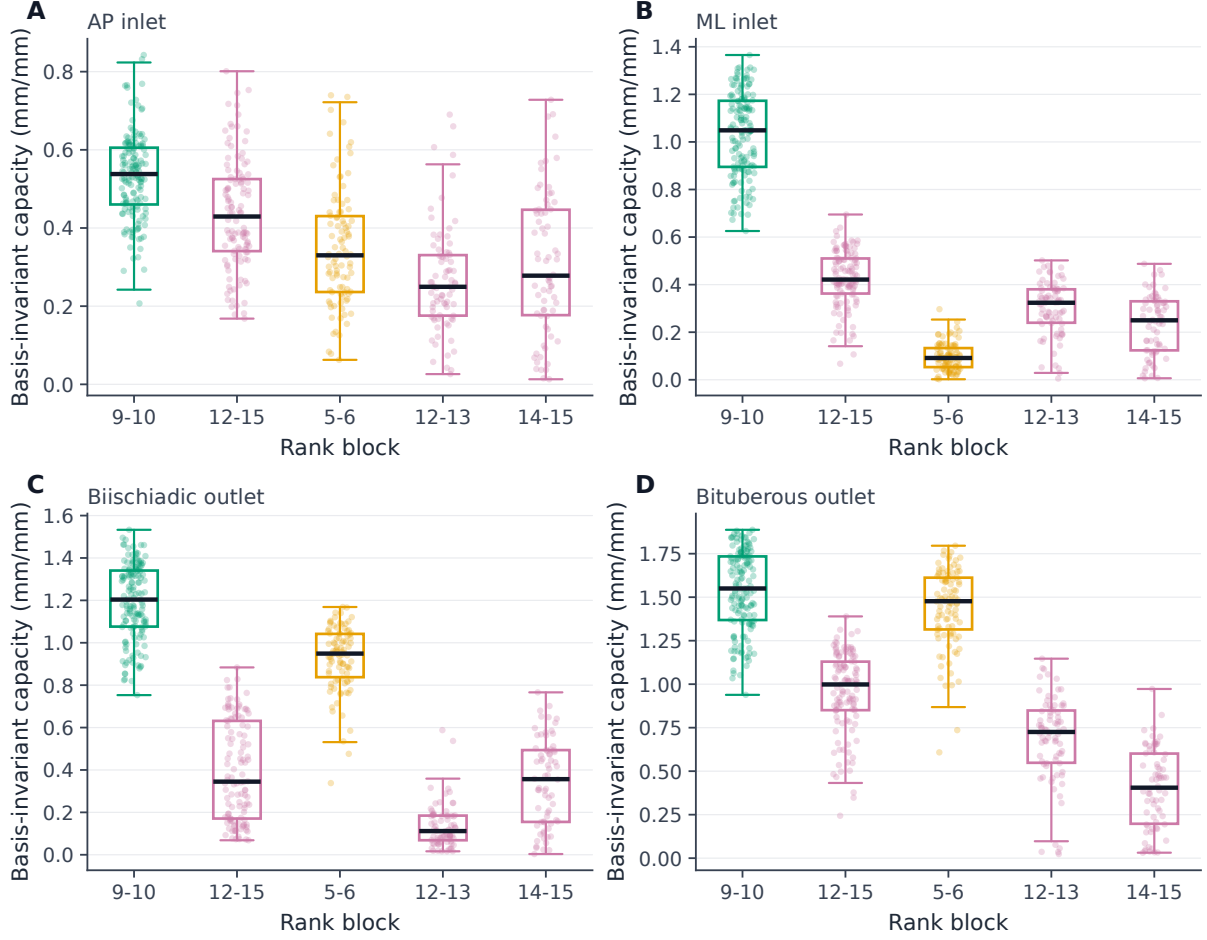


Figure 3: **Basis-invariant anatomical capacities across recurrent clustered modal subspaces.** Basis-invariant anatomical capacities (maximum anatomical diameter response per 1 mm maximum nodal displacement) for the five most prevalent rank blocks across the cohort ($N=278$): **(A)** anteroposterior (AP) inlet, **(B)** mediolateral (ML) inlet, **(C)** biischadic outlet, and **(D)** bituberous outlet. Boxes indicate median and interquartile range (IQR); whiskers extend to $1.5 \times \text{IQR}$.

the inlet-related signal disappears, but because it shifts inside the same two-dimensional block while the solver labels exchange ($p < 10^{-21}$ for both modes).

By contrast, the *subspace-level* AP capacity of the 9–10 block remains substantial across both groups (basis-invariant single-functional capacity $\kappa_{\text{AP},\infty}$: median 0.551 vs. 0.525 mm/mm; difference -0.026 mm/mm, 95% CI [-0.084, 0.015], ratio 0.95, $p = 0.030$, $r_{\text{rb}} = -0.21$; companion joint-SVD score: median 0.077 vs. 0.075 mm/mm, difference -0.002 mm/mm, 95% CI [-0.020, 0.012], $p = 0.840$). In sharp contrast to the massive, statistically decisive redistribution seen at the single-mode level ($p < 10^{-21}$), the subspace-level capacity exhibits only modest descriptive variation between non-swapped and swapped groups (difference -0.026 mm/mm, 95% CI [-0.084, 0.015], $p = 0.030$). Across the entire unselected cohort ($N=278$; 204 no-exchange vs. 74 exchange), rank 9 coupling shifts by -0.319 mm/mm ($p = 9.7 \times 10^{-34}$) and rank 10 shifts by +0.301 mm/mm ($p = 4.3 \times 10^{-34}$), whereas subspace-level single-functional capacity differs by only -0.006 mm/mm (the median difference is minimal with a 95% bootstrap CI of [-0.048, 0.019] spanning zero, and the Mann–Whitney test detects no significant shift across the full cohort, $p = 0.106$, in sharp contrast to the massive single-mode redistribution where $\Delta \approx 0.30$ – 0.32 mm/mm and $p < 10^{-33}$). What becomes unstable is the attachment of the function to one rank index rather than another, not the anatomical capability itself. Across-subject comparisons

of higher-order clustered modes phrased as though “mode 9” or “mode 10” were fixed entities across the cohort can therefore be misleading. For this regime, the biologically meaningful unit is a compact recurrent subspace—here the shared 9–10 block.

Load routing reinforces the same conclusion (Fig. 5). The 9–10 exchange is primarily a labelling instability with quantitative tuning, not a loss of the inlet-related program. Subspace rotation and the internal 9–10 gap vary across subjects, AP coupling shifts modestly with swap status, and the throughput of the 9–10 block under LAB₁–LAB₃ depends on morphology in a phase-specific way. For LAB₁, block throughput decreases with larger AP inlet diameter ($\rho_s = -0.48$, $p = 4.3 \times 10^{-17}$), whereas LAB₂ and LAB₃ show weaker and differently signed trends. These patterns support the interpretation that inlet-related mechanics remains associated with the 9–10 subspace even as anatomical variability redistributes its basis representation and modulates its phase-specific amplitude.

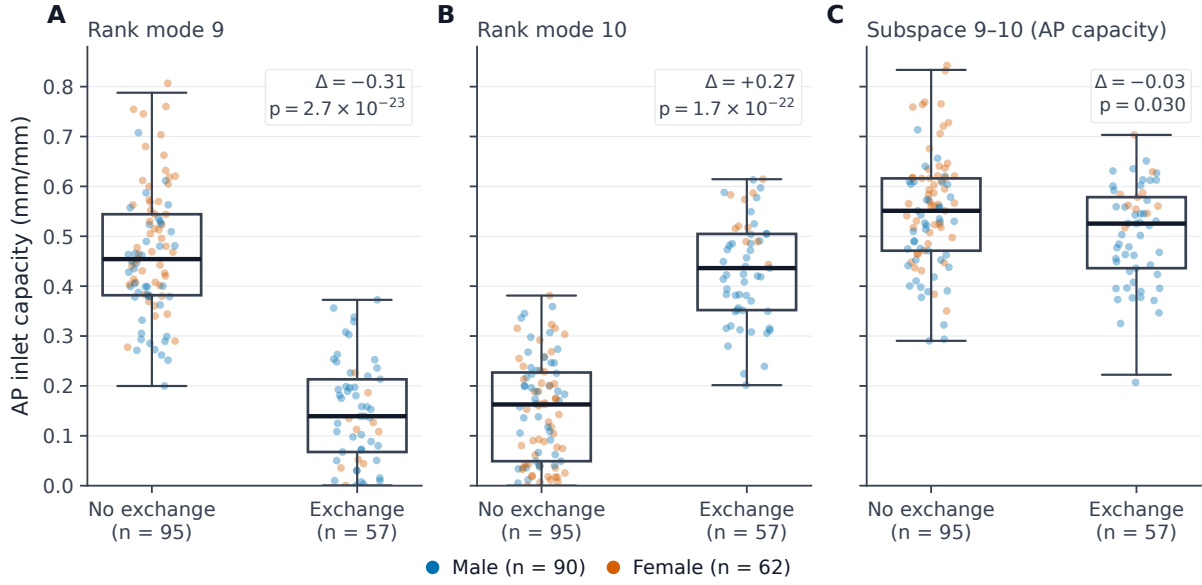


Figure 4: **Inlet coupling redistributes between rank labels within the stable 9–10 subspace.** Anteroposterior (AP) inlet coupling for rank mode 9 (A), rank mode 10 (B), and the basis-invariant single-functional AP capacity of the two-dimensional 9–10 subspace (C), stratified by $9 \rightleftharpoons 10$ label-exchange status. Points represent individual cluster-member subjects ($n=152$; no-exchange $n=95$, exchange $n=57$; male in blue, $n=90$; female in yellow, $n=62$). Horizontal lines indicate medians; boxes indicate interquartile range; whiskers indicate $1.5 \times \text{IQR}$. Coupling is expressed as diameter change per millimetre of maximum nodal displacement.

2.5 Standing and proxy loads distinguish backbone-dominated routing from higher-rank recruitment

Load routing reflects the same separation between conserved baseline support and selective higher-rank recruitment (Fig. 6, Table 2). Standing is dominated by the gap-protected low-rank backbone. Symmetric two-leg standing (SP2leg) is concentrated in mode 2 alone, which carries about 84.5% of the retained modal strain energy; single-leg standing (SP1leg) distributes energy across the full rotational triad (modes 1–3), reaching a top-3 cumulative share of about 92.1%. Within this linear setting, habitual load transfer therefore operates through a small, stable, low-dimensional program.

The proxy load family recruits higher-rank modes. LAB₁, bilateral mediolateral internal ring expansion (± 400 N outward), preferentially engages the rank 9–10 inlet-related subspace, which carries a mean energy share of about 0.51 and is led by the inlet-opening mode. LAB₂,

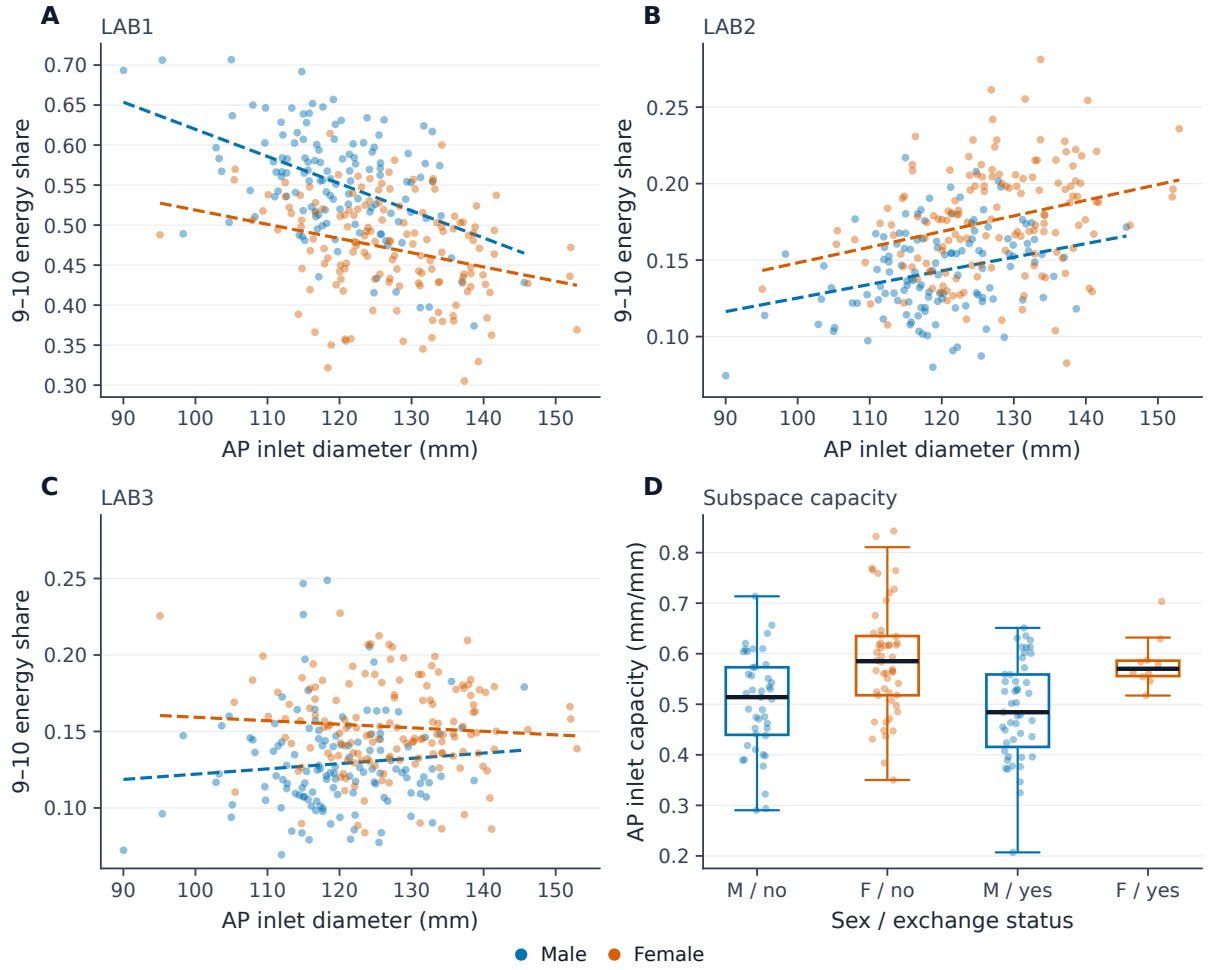


Figure 5: Morphology and loading modulate recruitment of the 9–10 modal block. (A–C) Modal strain-energy fraction carried by the 9–10 block as a function of AP inlet diameter for LAB₁ (bilateral internal ring expansion), LAB₂ (ischial distraction), and LAB₃ (outlet distraction). Blue dots and dashed lines indicate males; yellow dots and dashed lines indicate females. (D) Basis-invariant single-functional AP-inlet capacity of the 9–10 subspace stratified jointly by sex and label-exchange status (M/no, F/no, M/yes, F/yes). Boxes show median and IQR; whiskers extend to $1.5 \times \text{IQR}$.

bilateral ischial-tuberosity distraction, reroutes energy into the mid-rank block at modes 4–8, where the 5–6 outlet-related pair becomes especially important. LAB₃, anteroposterior outlet distraction, is hybrid, retaining sizeable low-rank contributions while also recruiting mid-rank and 9–10 pathways. Different loading geometries recruit different parts of the higher-rank spectrum, while continuing to use the backbone as common support.

Among the recurrent clustered regimes, 9–10 remains the clearest case because it combines high cohort prevalence, clear inlet-related interpretation, direct label-versus-subspace evidence (Fig. 4), and substantial recruitment under LAB₁. The 5–6 pair remains an outlet-related secondary contrast under LAB₂, whereas 12–15 is common but behaves, under the present load family, as a mixed higher-rank reserve rather than a single dominant pathway. Primary cohort routing is evaluated on the total static response; an incremental sensitivity check isolating $\delta \mathbf{u}_{\text{ext}} = \mathbf{K}_s^{-1} \mathbf{f}_{\text{ext}}$ on $N = 10$ representative subjects confirmed that removing baseline ligament pretension leaves modal dominance rankings and functional conclusions invariant (shifts $\leq 3.5\%$; Section 4.4). Additional load-specificity and sex-stratified summaries are reported in Appendix B and Appendix C.

These are linear comparative proxy loads, not simulations of labour. They do not model fetal contact, tissue remodelling, hormone-dependent softening, or nonlinear late-stage birth mechanics. Their value is comparative: they reveal which deformation programs are preferentially recruited under different loading geometries, not whether the model can predict clinical childbirth outcomes.

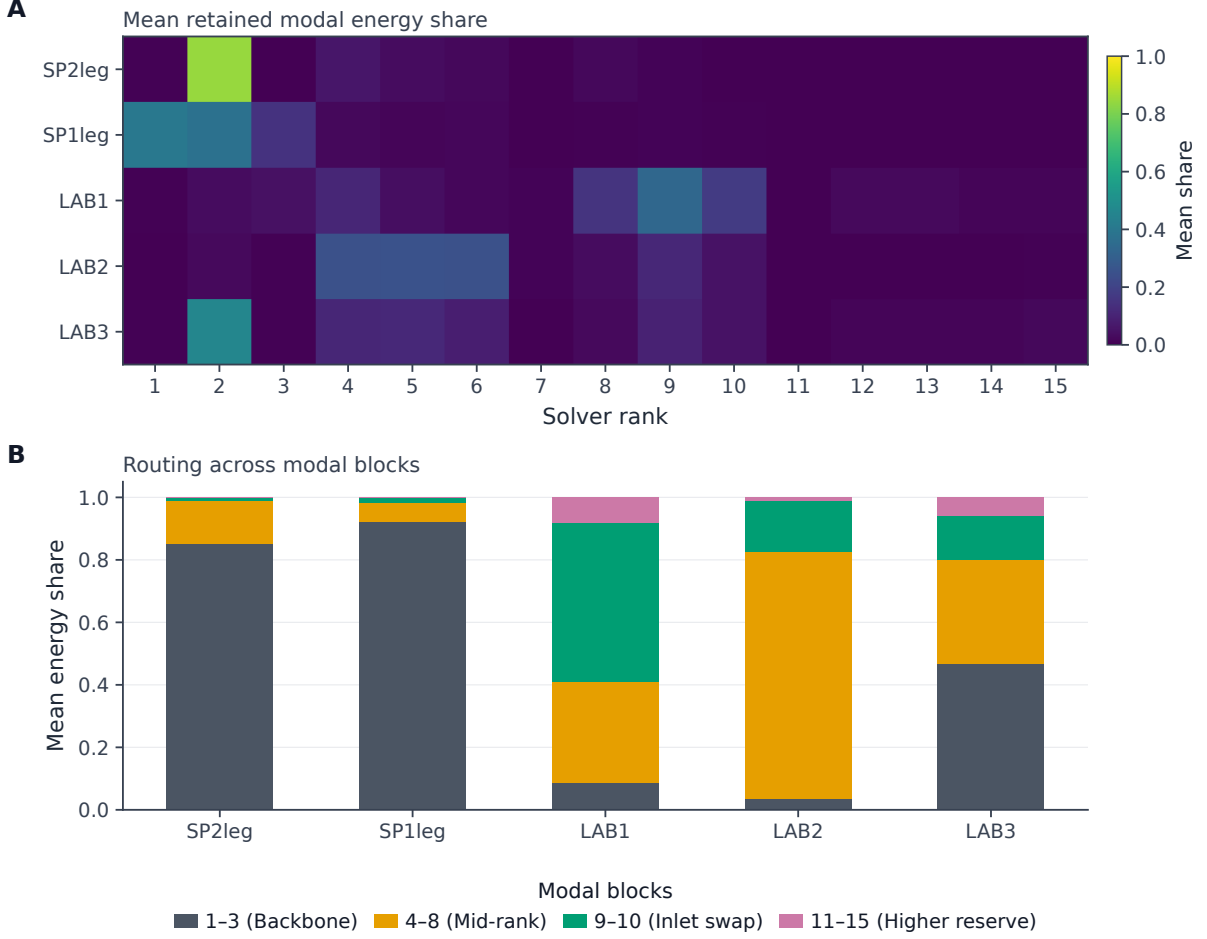


Figure 6: **Standing and proxy childbirth loads recruit distinct modal regimes.** **(A)** Heatmap of mean retained modal strain-energy share across the 15 lowest solver ranks for standing (SP2leg, SP1leg) and proxy childbirth loads (LAB₁–LAB₃). **(B)** Partition of retained modal strain energy across functional blocks: low-rank backbone (modes 1–3; blue), mid-rank regime (modes 4–8; orange), inlet cluster (modes 9–10; green), and higher-rank reserve (modes 11–15; purple). Standing concentrates in the backbone, whereas proxy loads recruit higher ranks in a task-specific pattern.

Table 2: **Dominant mode sets for the standing and proxy loading cases.** N_{80} denotes the smallest set of modes required to reach 80% cumulative mean energy fraction.

Load	Top mode	Top mode share (%)	Top-3 modes	Top-3 share (%)	N_{80}	Modes in 80% set	Cumulative share at N_{80} (%)
SP2leg	2 (ML rotation)	84.49	2,4,5	93.79	1	2	84.49
SP1leg	1 (CC rotation)	39.89	1,2,3	92.04	3	1,2,3	92.04
LAB ₁	9 (inlet opening)	33.49	9,10,8	66.12	5	9,10,8,4,3	81.48
LAB ₂	5	25.15	5,6,4	74.77	4	5,6,4,9	85.8
LAB ₃	2 (ML rotation)	45.81	2,5,4	68.09	5	2,5,4,9,6	85.8

2.6 Load-routing overlap distinguishes shared backbone support from selective clustered recruitment

Overlap analysis confirms that shared low-rank support and task-specific higher-rank routing are not the same thing. Both standing cases remain backbone-dominated, yet they are not identical: SP2leg reaches 80% cumulative throughput with mode 2 alone, whereas SP1leg requires the full triad modes 1–3. Among the proxy loads, LAB₁ is distinguished by its emphasis on the 9–10 inlet-related subspace, whereas LAB₂ and LAB₃ share greater mid-rank content, which clarifies why the 5–6 pair is an outlet-related secondary result rather than a side observation. Shared dominant modes and shared full routing profiles are therefore complementary but inequivalent descriptors. Population-scale spectral structure thus separates a conserved low-rank backbone from a selectively recruited higher-rank repertoire whose function is typically more stable at the subspace level than at the level of individual labels. Detailed pairwise overlap values are provided in Appendix C.

2.7 A minimal two-mode model reproduces the observed label–subspace dissociation

The organisation observed in the pelvic cohort admits a compact analytical interpretation in terms of the two- and three-degree-of-freedom models of Sec. 4.13. Fig. 7 shows a stochastic cohort in which a morphology coordinate μ modulates the detuning of a clustered pair on top of a gap-protected backbone, with additional cohort-level nuisance variability introduced through a compliant hidden pathway of scale η . The three model behaviours mirror the three empirical findings of the preceding subsections. First, the backbone mode keeps a stable rank across all subjects and all values of μ (Fig. 7A), matching the order-locked behaviour of modes 1–3 in the FE cohort. Second, as μ crosses the near-degenerate zone, the two cluster labels exchange their coupling to an inlet-like functional while the two-dimensional cluster subspace continues to carry the function (Fig. 7B), reproducing the label-versus-subspace split documented for the 9–10 pair (Fig. 4). Third, a standing-like load routes almost entirely through the backbone, while a task-like load recruits the clustered block, with the label-level split between its two modes moving between subjects but the block-level recruitment staying high (Fig. 7C), mirroring the backbone-versus-higher-rank contrast in the SP2leg/SP1leg and LAB₁–LAB₃ load cases. Increasing the hidden-pathway variability scale η at fixed morphology broadens the label-swap rate while leaving subspace coupling and task routing essentially unchanged (Fig. 7D), illustrating how cohort-level nuisance variability expected from unmodelled tissue heterogeneity can change which mode carries the function without changing whether the function is carried. The two-dimensional cluster in the toy model plays the same role as $\text{span}(Q_s)$ in the FE framework, and the morphology coordinate μ illustrates how a continuous anatomical shift can detune eigenvalues across an avoided crossing (minimum-gap) veering region. Importantly, our empirical permutation null test across the patient cohort (Table A6) did not show statistically significant localisation of label exchange along the AP diameter axis alone (all permutations yielded empirical $p > 0.05$, designated “NO” across test cases). This negative null test is scientifically revealing: it demonstrates that the toy model serves as a conceptual and mechanistic illustration of parameter-driven detuning and eigenvalue veering across a near-degenerate zone, rather than proof of one-dimensional anatomical causality. In real pelvic anatomy, mode exchange is governed by the concerted, multi-dimensional variation of the complete 3D ring geometry and boundary restraints, not by an isolated 1D diameter. This minimal model therefore formalises why the interpretive unit must shift from *mode label* to *cluster subspace* precisely in the regime in which spectral gaps collapse, and why this shift preserves anatomical deformation specificity without requiring one-dimensional geometric determinism.

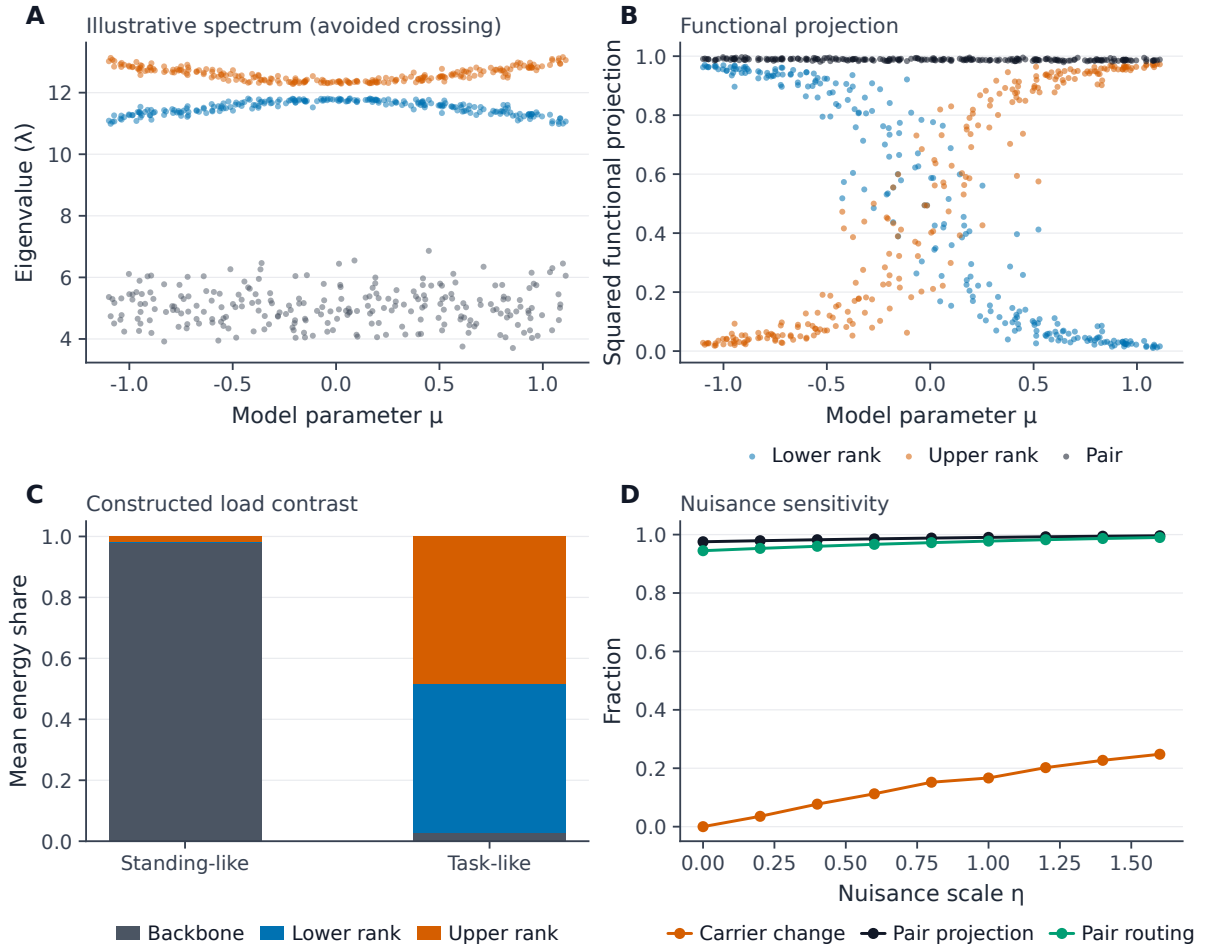


Figure 7: **Minimal analytical model illustrates label exchange within a stable subspace.** A stochastic cohort of the analytical model with morphology parameter μ and hidden-pathway scale η . **(A)** Spectrum: a separated backbone mode and a nearby eigenvalue pair undergoing near-degeneracy around $\mu \approx 0$. **(B)** Functional projection: coupling shifts between individual rank labels across the transition zone while total coupling within the pair remains high. **(C)** Load routing: standing-like load concentrates in the backbone, whereas task-like load recruits the clustered block with a variable between-label split. **(D)** Effect of hidden-pathway variability scale η : label-exchange rate increases with η , whereas subspace coupling and task routing remain stable. Model derivations and simulation parameters are detailed in Methods.

3 Discussion

Subspace stability resolves the tension between anatomical variability and functional robustness

This pelvic cohort demonstrates that anatomical deformation programs can remain robust across substantial skeletal variability without requiring rigid one-to-one conservation of individual spectral ranks. Habitual load transfer is concentrated in a low-rank backbone, whereas higher-order anatomical deformations remain remarkably stable within compact recurrent subspaces. The essential biological insight is not merely that spectral labels can become unstable, but that anatomical variation redistributes deformation within a compact local repertoire while maintaining the anatomical capability itself.

This distinction separates three objects often conflated in comparative biomechanics: anatomical shape, deformation pathway, and spectral rank. Shape varies continuously across individ-

uals; deformation pathways describe physical kinematic motions of the ring; and spectral rank merely sorts these pathways by ascending stiffness. When analysis is restricted to fixed eigenvector ranks, subject-to-subject variability can easily be mistaken for functional loss or chaotic instability. By evaluating the system through invariant modal subspaces, we demonstrate that mechanical meaning is not lost; rather, the appropriate unit of biological organisation shifts from the isolated eigenmode to a compact, recurrent subspace.

A concrete practical implication follows for population biomechanics: when comparing inlet compliance across individuals using rank mode 9 alone, subjects who undergo a $9 \rightleftharpoons 10$ label exchange (57 of 152 cluster members, 37.5%) exhibit an apparent collapse of AP-inlet coupling (from median 0.454 down to 0.139 mm/mm, an apparent drop of $\approx 69\%$), falsely suggesting a severe impairment of ring dilation. In reality, evaluating the shared two-dimensional 9–10 subspace reveals that the anatomical opening capability is substantially more stable across both groups (basis-invariant single-functional capacity: 0.551 vs. 0.525 mm/mm; joint canonical coupling: 0.077 vs. 0.075 mm/mm). Adopting a subspace framework directly resolves false-negative biomechanical classifications that arise purely from mode tracking across near-degenerate regimes.

Functional organization across spectral scales: Conserved backbone versus flexible higher-rank reserves

The low-rank backbone represents the order-locked regime. Modes 1–3 remain spectrally separated, rarely reorder across the cohort ($< 0.4\%$ swap rate), and dominate habitual standing loads (carrying 84.5% of retained modal strain energy in symmetric two-leg stance and 92.0% across the triad in single-leg stance). Because eigenvalues λ_k quantify modal stiffness, these lowest-eigenvalue modes represent the highest-compliance baseline deformation pathways of the pelvic ring. Within the present linear framework, this supports the view that the low-order kinematic pathways of the pelvic ring are governed by a conserved, mechanically accessible baseline program. We do not infer an evolutionary optimisation principle from these data alone, but the result is physically consistent with biological robustness being anchored in a compact, gap-protected foundation [5, 6].

The higher ranks reveal an organisation characterised by recurrent clustered blocks. The rank 9–10 pair exemplifies this regime: inlet-related dilation is not reliably attached to a single rank label across the cohort, but the shared two-dimensional subspace retains substantial AP-inlet coupling and is strongly recruited under parturition-aligned proxy loads (LAB₁). Rather than positing an unverified evolutionary selection for structural degeneracy, our results provide a physical interpretation compatible with mechanics: in complex ring structures with near-degenerate eigenvalue spectra, spectral clustering arises naturally as a consequence of three-dimensional ring geometry and material heterogeneity, enabling structurally distinct internal pathways to provide equivalent functional deformations without requiring exact mathematical symmetry [6, 7, 15]. In parallel, the 5–6 block forms an outlet-related subspace with strong biischadic coupling, whereas the broader 12–15 family behaves as a higher-order reserve with diffuse canonical coupling.

Implications for the obstetrical dilemma and population biomechanics

The human pelvic ring has long been studied through the framework of the “obstetrical dilemma” [1, 2], which posits an evolutionary tension between the structural rigidity necessary for efficient bipedal locomotion and the internal dimensions required for the birth of large-brained neonates. Our modal analysis provides a mechanistic perspective on how these competing functional demands are partitioned across spectral scales. Locomotor loads (such as standing and single-leg support) are absorbed primarily by the low-eigenvalue baseline modes (modes 1–3), which provide the natural compliance pathways for habitual weight-bearing. Conversely, parturition-aligned loading conditions do not simply engage this baseline triad; instead, they selectively re-

cruit higher-rank clustered modes—most prominently the inlet-dilating 9–10 subspace—whose kinematically specialized deformation fields directly widen birth-canal diameters.

Furthermore, our minimal analytical model (Sec. 2.7) demonstrates that cohort-level nuisance variability—such as that arising from unmodelled ligamentous laxity, pelvic floor compliance, or subtle morphological detuning—can easily induce label veering between nearby modes. Crucially, as shown in Fig. 7D, increasing the scale of hidden-pathway variability (η) accelerates the rate of label exchange between individual modes without degrading subspace-level functional coupling or load throughput. This indicates that the pelvic ring can tolerate individual-level variations in soft-tissue stiffness and shape while reliably preserving substantial capacity for inlet and outlet expansion.

Methodological scope, physiological limitations, and future directions

What generalises from this study is the logic for testing robustness, not the exact pelvic cluster layout. Combining spectral gap structure, subspace comparison via Grassmann distance, anatomically defined functionals, and modal strain-energy routing makes it possible to evaluate whether function is maintained through stable individual pathways or through stable subspaces in any anatomically variable, load-bearing biological system. The specific rank partition observed here should not be transferred uncritically to other skeletal assemblies or altered physiological states, nor should every recurrent cluster be assumed to map cleanly onto a single anatomical degree of freedom.

Several limitations bound the physiological interpretation of these findings. First, the present analysis is linearised, evaluating the initial tangent stiffness operator of the pelvic ring. Child-birth is a complex, contact-driven, large-deformation process accompanied by progressive viscoelastic relaxation and hormone-induced ligament softening (principally via relaxin and progesterone) [13, 16–18]. The linear modal spectrum characterises the deformation repertoire immediately available before large-scale joint separation occurs; it does not simulate the descent of the fetal head or predict clinical labour outcomes [19]. Second, the cohort was assembled from clinical CT scans with a mean subject age of ≈ 54 years, which may not fully represent the pelvic morphology of primiparous women. Third, while baseline cartilage and ligament stiffnesses were calibrated against literature values, subject-specific ligament properties were not available. Our material uncertainty analysis employs first-order Taylor series propagation rather than full non-linear Monte Carlo sampling, which is computationally prohibitive across 278 high-resolution 3D finite-element meshes; while this local linear approximation indicates localized sensitivity trends and covariance-aware gap variances, it represents a first-order approximation that does not capture higher-order non-linear parameter interactions or full non-linear distributions. Nonetheless, our sensitivity analyses (Appendix D) indicate that realistic material uncertainty raises overall spectral variability without reorganising the underlying gap architecture or dismantling the clustered regimes. Extending this subspace framework to include geometrically nonlinear deformation, full stochastic sampling, and individualised soft-tissue kinematics represents a promising direction for future population biomechanics.

4 Materials and methods

The Methods below state the computational framework in self-contained form. The core objects are (i) a pullback-assembled linear-elasticity problem on a shared reference domain, (ii) a subject-specific generalised eigenproblem, (iii) a cross-subject pairing on a common mass metric, (iv) a gap-based detector for recurrent clustered regimes, (v) invariant-subspace comparison via principal angles and the Grassmann distance, and (vi) basis-invariant anatomical functionals that supply physically interpretable coordinates inside otherwise ambiguous clustered subspaces. Wherever possible, we use notation that separates three distinct objects: the solver *rank* index j

(the j -th smallest eigenvalue for each subject), the paired *reference* label $i = \pi_s^{-1}(j)$ (a cross-subject identifier via inverse assignment), and the *anatomical* functional ℓ (a linear observable defined on the template mesh).

4.1 Ethics statement

This study analysed a previously published, anonymised imaging-derived dataset and its associated precomputed finite-element repository (BoneDat) [3]. No new human imaging or interventions were performed for this work. According to the BoneDat descriptor, use of the source imaging data was approved by the Ethics Committee of the University Hospital Hradec Kralove (reference 202411IO3P) with waiver of informed consent; we refer readers to the dataset publication for full ethical and governance details.

4.2 Cohort and geometry normalisation

We analysed a precomputed set of $N=278$ CT-derived pelvic finite-element models assembled on a common reference domain ($n_F=150$, $n_M=128$; age 16–91 yr, mean 54 ± 19). No new cohort was recruited. The source repository provides standardised pelvic geometries suitable for in silico analysis [3]. The reference pelvic domain Ω_{ref} was discretized using a conforming volumetric mesh comprising $\approx 410\,000$ linear four-node tetrahedral elements (Lagrange P_1 , totaling $\approx 1.2 \times 10^6$ displacement degrees of freedom), with conforming interfaces between the bony segments (sacrum, left and right ossa coxae) and the joint cartilage domains (bilateral sacroiliac articular cartilage and the pubic symphysis disc). A diffeomorphic registration $\chi_s : \Omega_{\text{ref}} \rightarrow \Omega_s$ with deformation gradient $F_s(X) = \nabla_X \chi_s(X)$ and Jacobian $J_s = \det F_s > 0$ maps the template domain Ω_{ref} onto each subject domain Ω_s . The pipeline standardises bulk pelvic size so that the main population analysis isolates shape-driven and material-driven variability rather than absolute scale. Figure 8 summarises the anatomical dimensions, ligament representation, and boundary-condition layout used to define the pelvic models analysed here.

4.3 Pullback-assembled linear elasticity on a common reference domain

Let $\Omega_{\text{ref}} \subset \mathbb{R}^3$ be the template domain and Ω_s the subject domain, related by a fixed diffeomorphic registration $x = \chi_s(X)$ with deformation gradient $F_s(X) = \nabla_X \chi_s(X)$ and Jacobian $J_s(X) = \det F_s > 0$. Writing $\chi_s(X) = X + \varphi_s(X)$, so $F_s = I + \nabla_X \varphi_s$, small-strain linear elasticity is solved on Ω_s but integrated on Ω_{ref} through the standard change of variables [19, 20]. For trial and test fields $\hat{\mathbf{u}}, \hat{\mathbf{v}} \in V \subset [H^1(\Omega_{\text{ref}})]^3$, the physical strain tensor at $x = \chi_s(X)$ is

$$\boldsymbol{\varepsilon}_x(\mathbf{u}) = \text{sym}(\nabla_x \mathbf{u}) = \text{sym}(\nabla_X \hat{\mathbf{u}} F_s^{-1}), \quad \boldsymbol{\sigma}_x = C_{\text{phys}}(x) : \boldsymbol{\varepsilon}_x(\mathbf{u}), \quad (1)$$

and the pullback-assembled elastic bilinear form reads

$$a(\hat{\mathbf{u}}, \hat{\mathbf{v}}) = \int_{\Omega_{\text{ref}}} \left[C_{\text{phys}}(\chi_s(X)) : \text{sym}(\nabla_X \hat{\mathbf{u}} F_s^{-1}) \right] : \text{sym}(\nabla_X \hat{\mathbf{v}} F_s^{-1}) J_s \, dX. \quad (2)$$

Here F_s and J_s arise only from the fixed image registration; no finite-strain mechanics is implied. The mass bilinear form inherits the geometric volume factor,

$$m(\hat{\mathbf{u}}, \hat{\mathbf{v}}) = \int_{\Omega_{\text{ref}}} J_s(X) \hat{\mathbf{u}} \cdot \hat{\mathbf{v}} \, dX, \quad (3)$$

defining an L^2 volume inner product weighted by local volumetric dilation rather than physical dynamic mass density. The resulting generalised eigenvalues $\lambda_k = \boldsymbol{\phi}_k^\top K_s \boldsymbol{\phi}_k / (\boldsymbol{\phi}_k^\top M_s \boldsymbol{\phi}_k)$ quantify modal stiffness (or squared pseudo-frequencies), while the reciprocal eigenvalues λ_k^{-1} describe modal compliance rather than physical resonance frequencies. The *unmapped* reference mass

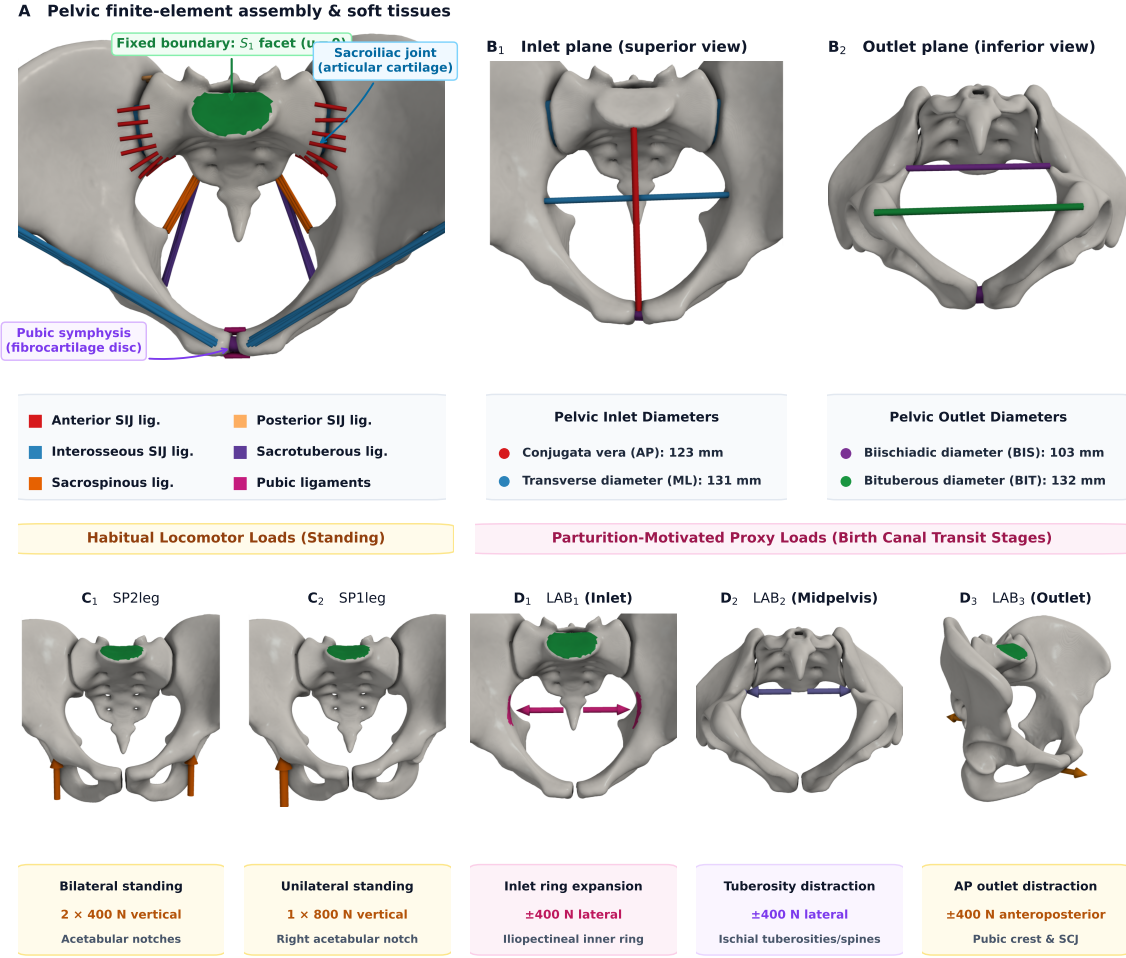


Figure 8: Pelvic finite-element assembly, anatomical diameters, boundary conditions, and static load configurations. (A) Complete 3D pelvic finite-element model comprising the sacrum, bilateral ossa coxae, sacroiliac joint (SIJ) articular cartilage (cyan), and pubic symphysis fibrocartilage disc (purple). Eleven discrete anatomical ligament bundles (anterior SIJ, posterior SIJ, interosseous SIJ, sacrospinous, sacrotuberous, and pubic ligaments) are explicitly represented. The superior articular surface of the sacrum (S_1 facet, green) is fixed against rigid-body displacement ($\mathbf{u} = \mathbf{0}$) via surface penalty. (B) True planar normal projections for anatomical pelvimetric diameters: (B₁) pelvic inlet plane showing Conjugata vera (true conjugate, red, 123 mm) and transverse diameter (blue, 131 mm); (B₂) pelvic outlet plane (inferior/perineal projection) showing biischadic (interspinous) diameter (purple, 103 mm) and bituberous diameter (green, 132 mm). (C) Habitual locomotor load cases representing upright stance: (C₁) bilateral standing (SP2leg, 2 × 400 N vertical joint reaction forces applied at left and right acetabular notches); (C₂) unilateral single-leg stance (SP1leg, 1 × 800 N vertical joint reaction force applied at the right acetabulum). (D) Parturition-motivated proxy load cases representing stages of fetal transit through the pelvic canal: (D₁) inlet ring expansion during head engagement (LAB₁, ±400 N lateral expansion forces applied at the iliopectineal inner ring contacts); (D₂) midpelvis transit (LAB₂, ±400 N lateral distraction forces applied across the ischial tuberosities/spines); (D₃) outlet clearance (LAB₃, ±400 N anteroposterior distraction consisting of anterior traction on the pubic crest and posterior traction on the sacrococcygeal joint).

matrix M_{ref}^0 is obtained from (3) with $J \equiv 1$ and provides a shared, registration-independent inner product across subjects.

Material model. The physical elasticity tensor $C_{\text{phys}}(x)$ is isotropic and heterogeneous. For bone, the CT-derived hydroxyapatite density $\rho(x)$ is converted to ash density $\rho_{\text{ash}} = (\rho + 0.09)/1.14$ and mapped to a Young's modulus $E(x)$ via the Keller [21] piecewise power law,

$$E(x) = \begin{cases} 2398 \text{ MPa}, & \rho_{\text{ash}}(x) \leq 0.486, \\ \alpha \rho_{\text{ash}}(x)^\beta, & \rho_{\text{ash}}(x) > 0.486, \end{cases} \quad \alpha = 10\,200, \beta = 2.0, \quad (4)$$

with Poisson's ratio $\nu = 0.3$. This family of density–modulus relations brackets published calibrations spanning the classical Carter–Hayes exponent [22] and the site-dependent regressions of Morgan et al. [23]. Cartilage (sacroiliac joints and pubic symphysis) is homogeneous linear elastic with $E = 1.0 \text{ MPa}$ and $\nu = 0.45$. These choices are consistent with structural adaptation and architecture–stiffness arguments in bone biomechanics [24–27] and with the broader sacroiliac and pubic-symphysis mechanics literature used to motivate linearised pelvic models [12–14].

Ligaments in pullback coordinates. Ligament families are represented as polyline bundles mapped from template nodes X_i to physical points $x_i = \chi_s(X_i)$. For each segment (X_i, X_j) define the physical unit tangent $\mathbf{e} = (x_j - x_i)/\|x_j - x_i\|$ and length $L = \|x_j - x_i\|$. The total bundle stiffness and pretension are distributed across N_{seg} segments, yielding per-segment $k_{\text{seg}} = K_{\text{bundle}}/N_{\text{seg}}$ and $T_{\text{seg}} = T_{\text{bundle}}/N_{\text{seg}}$. Each segment contributes an axial stiffness block

$$W = k_{\text{seg}}(\mathbf{e}\mathbf{e}^\top), \quad K_{ij}^{\text{ax}} = \begin{bmatrix} +W & -W \\ -W & +W \end{bmatrix}, \quad (5)$$

and, in the presence of pretension $T_{\text{seg}} \geq 0$, a geometric (prestress) stiffness

$$K_{ij}^{\text{geo}} = \frac{T_{\text{seg}}}{L}(\mathbf{I} - \mathbf{e}\mathbf{e}^\top) \succeq 0, \quad (6)$$

together with paired nodal forces $\pm T_{\text{seg}}\mathbf{e}$ on the right-hand side. The full assembled tangent stiffness evaluated at the prestressed reference state is

$$K_s = K_{\text{elas},s} + K_{\text{lig},s} + K_s^{\text{geo}}(T) + K_{\Gamma,s}, \quad f = f_{\text{ext}} + f_{\text{pret}}(T), \quad (7)$$

where geometric stiffness $K_s^{\text{geo}}(T)$ incorporates the stabilizing effect of initial ligament pretension T . In our linearised modal framework, incremental perturbations are evaluated directly about this prestressed configuration without requiring iterative non-linear equilibrium updates. The term $K_{\Gamma,s}$ implements Dirichlet penalties on Γ_{ref} , $a_\Gamma(\hat{\mathbf{u}}, \hat{\mathbf{v}}) = \gamma \int_{\Gamma_{\text{ref}}} \hat{\mathbf{u}} \cdot \hat{\mathbf{v}} J_{\Gamma,s} dS_X$ with $J_{\Gamma,s} = J_s \|F_s^{-T} N\|$ and $\gamma = \alpha_\Gamma \bar{E}/h$, $\alpha_\Gamma \in [10^3, 10^5]$. A decade sweep of γ left the reported cluster classification unchanged.

Force-controlled loading. Neumann terms are pulled back to Γ_{ref} so that physical resultants are invariant under χ_s :

$$\int_{\Gamma_s} \mathbf{t}_{\text{phys}} \cdot \mathbf{v} dS_x = \int_{\Gamma_{\text{ref}}} J_{\Gamma,s}(X) \mathbf{t}_{\text{phys}}(\chi_s(X)) \cdot \hat{\mathbf{v}}(X) dS_X. \quad (8)$$

All load cases in this study are defined by a total force $\mathbf{F}$ on a tagged physical region; the applied traction is $\mathbf{t}_{\text{phys}} = \mathbf{F}/A_{\text{phys}}$ with $A_{\text{phys}} = \int_{\Gamma_{\text{ref}}} J_{\Gamma,s} dS_X$, so that $\int_{\Gamma_s} \mathbf{t}_{\text{phys}} dS_x = \mathbf{F}$ for every subject.

4.4 Standardised load cases

Loads were applied as force-controlled Neumann tractions via (8) so that every subject experienced the same integrated force vector regardless of physical surface area. Two standing cases were considered: symmetric two-leg stance (SP2leg; 400 N per acetabulum, +CC)

and one-leg stance (SP1leg; 800 N on the stance acetabulum, +CC). The 400 N per hip in SP2leg is a conservative split of head–arms–trunk mass onto both acetabula based on classical segment-inertia data [28], and the 800 N one-leg value is well below in-vivo hip-implant peaks of $2.5\text{--}3.5\times\text{BW}$ recorded during single-leg support and gait [29, 30]. Three proxy load cases probed parturition-relevant routing without modelling contact mechanics: LAB₁ applied bilateral mediolateral internal ring expansion forces (± 400 N outward), LAB₂ applied bilateral ischial-tuberosity distraction (± 400 N), and LAB₃ applied anteroposterior outlet distraction between the pubis insertion ($-AP$, 400 N) and the sacrococcygeal joint ($+AP$, 400 N). These loads act as standardized comparative probes of pelvic structural compliance rather than clinical simulations of labour [17, 18]. Static finite-element solutions solve the linear equilibrium equation $K_s \mathbf{u} = \mathbf{b}$ with

$$\mathbf{b} = \mathbf{f}_{\text{ext}} + \mathbf{f}_{\text{pret}} \quad (9)$$

for the primary total-response analysis, where $\mathbf{f}_{\text{pret}}$ represents baseline ligament pretension (20 N per ligament bundle) and $\mathbf{f}_{\text{ext}}$ represents external load. Primary cohort-wide modal strain-energy fractions (Fig. 6, Table 2) are computed directly from these total displacement fields $\mathbf{u}$. By linearity, total displacement decomposes as $\mathbf{u} = \mathbf{u}_{\text{pret}} + \delta\mathbf{u}_{\text{ext}}$, where $\delta\mathbf{u}_{\text{ext}} = K_s^{-1}\mathbf{f}_{\text{ext}}$ is the incremental external load response. Under external load scaling $\mathbf{f}_{\text{ext}} \rightarrow c\mathbf{f}_{\text{ext}}$, incremental displacements scale proportionally ($\delta\mathbf{u}_{\text{ext}} \rightarrow c\delta\mathbf{u}_{\text{ext}}$), such that modal expansion coefficients $q_i = \phi_i^\top M_s \delta\mathbf{u}_{\text{ext}}$ scale by c , modal energies $E_i = \frac{1}{2}\lambda_i q_i^2$ scale by c^2 , and the normalized modal strain-energy fractions $f_i = E_i / \sum_j E_j$ of the incremental response are strictly scale-invariant. To verify that baseline pretension does not confound modal load routing, an incremental sensitivity analysis subtracting $\mathbf{u}_{\text{pret}}$ was performed on a representative subset of $N = 10$ subjects (5 exchange, 5 non-exchange across all 5 loads): top recruited modes matched in 100% of proxy cases and 90% of standing cases, functional block energy shifts were minor ($\leq 3.5\%$; Modes 9–10 inlet energy increased from 71.7% to 75.2%), and dominant set sizes (N_{80}) were identical, confirming that the primary total-response routing faithfully captures external load recruitment without requiring full-cohort re-simulation.

4.5 Generalised eigenproblem, rigid-mode filtering, and cross-subject pairing

For each subject s we assembled (K_s, M_s) on Ω_{ref} via (7) and (3) and solved the generalised eigenproblem

$$K_s \phi_k = \lambda_k M_s \phi_k, \quad \phi_i^\top M_s \phi_j = \delta_{ij}, \quad (10)$$

using a Krylov–Schur solver (SLEPc) with shift-invert spectral transform and MUMPS direct factorisation, extracting the $N_{\text{ev}} = 15$ lowest non-rigid eigenpairs. Rigid modes (numerically $\lambda < 10^{-8}$) were filtered. For cross-subject comparison we used the mass-weighted modal assurance criterion on the shared reference metric,

$$\text{MAC}_{M_0}(\mathbf{x}, \mathbf{y}) = \frac{|\mathbf{x}^\top M_{\text{ref}}^0 \mathbf{y}|^2}{(\mathbf{x}^\top M_{\text{ref}}^0 \mathbf{x})(\mathbf{y}^\top M_{\text{ref}}^0 \mathbf{y})} \in [0, 1], \quad (11)$$

and solved the Hungarian optimal-assignment problem on $-\text{MAC}_{M_0}$ over the full $N_{\text{ev}} \times N_{\text{ev}}$ matrix [31], yielding an assignment bijection $\pi_s : \{1, \dots, N_{\text{ev}}\} \rightarrow \{1, \dots, N_{\text{ev}}\}$, where $\pi_s(i_{\text{ref}}) = j_{\text{rank}}$ maps reference identifier i_{ref} to its rank position in subject s , and the inverse permutation $\pi_s^{-1}(j_{\text{rank}}) = i_{\text{ref}}$ assigns reference identity to each solver rank. Entries with $\text{MAC}_{M_0} < 0.05$ were flagged as unpaired. Rank index j (solver order) is required for gap statistics and cluster detection; the inverse assignment $\pi_s^{-1}(j)$ maps each solver mode back to its reference identity and is used only for anatomical interpretation, preserving the distinction between rank order and pathway identity emphasised in the Introduction.

4.6 Spectral gap structure and clustered-regime detection

For each subject with ordered eigenvalues $\lambda_1 \leq \dots \leq \lambda_{N_{\text{ev}}}$, relative consecutive gaps are

$$\text{gap}_{\text{in}}(i) = \frac{\lambda_{i+1} - \lambda_i}{\lambda_i}, \quad i = 1, \dots, N_{\text{ev}} - 1. \quad (12)$$

A clustered regime $[i..j]$ is defined as the maximal contiguous rank block starting at i with internal gaps below threshold, $\text{gap}_{\text{in}}(k) < \varepsilon_{\text{in}}$ for $k = i, \dots, j-1$. The threshold is set from data as the 25th percentile of the pooled gap_{in} distribution across the cohort, yielding $\varepsilon_{\text{in}} \approx 0.13$; sensitivity to the percentile choice is reported in Appendix A. External separation is quantified by

$$\text{sep}_{\text{out}}([i..j]) = \min\left(\frac{\lambda_i - \lambda_{i-1}}{\lambda_i}, \frac{\lambda_{j+1} - \lambda_j}{\lambda_j}\right), \quad (13)$$

defined for interior clusters ($i > 1$ and $j < N_{\text{ev}}$) where both adjacent neighbours exist. For boundary blocks reaching modal truncation (such as the 12–15 cluster with $N_{\text{ev}} = 15$), the upper neighbour is uncomputed and two-sided separation is formally undefined. Exact mathematical multiplicity ($\lambda_i = \lambda_{i+1}$) was not observed under the operational numerical tolerance $\text{gap}_{\text{in}} \leq \varepsilon_{\text{deg}} = 10^{-6}$ in any subject. No obvious spatial symmetry was identified that would enforce repeated eigenvalues, although accidental parameter-dependent crossings are not mathematically excluded; nonetheless, all regimes detected by (12) are genuine near-degenerate clusters, for which small perturbations can drive rapid internal basis rotation and label exchange [9, 10].

4.7 Invariant subspaces: principal angles, Grassmann distance, and Procrustes alignment

For each clustered rank block of dimension r , the corresponding subject-specific subspace was represented by an M_{ref}^0 -orthonormal basis

$$Q_s \in \mathbb{R}^{n \times r}, \quad Q_s^\top M_{\text{ref}}^0 Q_s = I_r. \quad (14)$$

With a fixed reference basis Q_{ref} , define the overlap matrix

$$S = Q_{\text{ref}}^\top M_{\text{ref}}^0 Q_s \in \mathbb{R}^{r \times r}, \quad S = U \Sigma V^\top, \quad \cos \theta_k = \sigma_k(S), \quad (15)$$

with singular values σ_k yielding the principal angles $\theta_k \in [0, \pi/2]$ between the two subspaces [32]. The Grassmann distance is

$$d_G(Q_{\text{ref}}, Q_s) = \|\theta\|_2 = \left(\sum_{k=1}^r \theta_k^2\right)^{1/2}. \quad (16)$$

When a stable basis representation inside a clustered subspace is required, we apply an orthogonal Procrustes rotation [33],

$$\tilde{Q}_s = Q_s (V U^\top), \quad (17)$$

which minimises the M_{ref}^0 -weighted distance $\|Q_{\text{ref}} - \tilde{Q}_s R\|_{M_{\text{ref}}^0}$ over orthogonal matrices $R \in O(r)$ while preserving the subspace $\text{span}(Q_s)$.

All quantitative biological comparisons use basis-invariant scalars derived from (15)–(16) or from the canonical-basis construction below.

Degeneracy vs. near-degenerate mixing. Exact mathematical multiplicity requires $\lambda_i = \lambda_{i+1}$ (operationalized numerically as $\text{gap}_{\text{in}}(k) \leq \varepsilon_{\text{deg}}$ for all k in the block, with $\varepsilon_{\text{deg}} \ll \varepsilon_{\text{in}}$); individual eigenvectors inside a degenerate subspace are then not unique, and any unitary rotation inside $\text{span}(Q_s)$ is an equally valid basis. When $\varepsilon_{\text{deg}} < \text{gap}_{\text{in}}(k) < \varepsilon_{\text{in}}$ the eigenvalues are close but distinct; the eigenvectors are formally unique, yet small perturbations can rotate the internal basis substantially while $\text{span}(Q_s)$ itself remains remarkably stable. This is the regime in which label exchange occurs without loss of the block-level mechanical content.

4.8 Canonical anatomical functionals inside clustered subspaces

To interpret clustered subspaces biologically, we use basis-invariant linear functionals defined by anatomical landmark pairs. On the reference template mesh, five key obstetric and locomotor diameters are defined:

- (i) **AP inlet** (s_{AP}): Promontory (X_{prom}) to superior pubic symphysis (X_{pub}), with unit direction $\mathbf{e}_{\text{AP}} = (X_{\text{pub}} - X_{\text{prom}}) / \|X_{\text{pub}} - X_{\text{prom}}\|$;
- (ii) **ML inlet** (s_{ML}): Widest transverse diameter across left and right iliopectineal lines ($X_{\text{left}}, X_{\text{right}}$), with unit direction $\mathbf{e}_{\text{ML}} = (X_{\text{right}} - X_{\text{left}}) / \|X_{\text{right}} - X_{\text{left}}\|$;
- (iii) **Biischiadic outlet** (s_{BIS}): Left to right ischial spine ($X_{\text{sp,left}}, X_{\text{sp,right}}$), with unit direction $\mathbf{e}_{\text{BIS}} = (X_{\text{sp,right}} - X_{\text{sp,left}}) / \|X_{\text{sp,right}} - X_{\text{sp,left}}\|$;
- (iv) **Bituberous outlet** (s_{BIT}): Left to right inferior ischial tuberosity ($X_{\text{tub,left}}, X_{\text{tub,right}}$), with unit direction $\mathbf{e}_{\text{BIT}} = (X_{\text{tub,right}} - X_{\text{tub,left}}) / \|X_{\text{tub,right}} - X_{\text{tub,left}}\|$;
- (v) **AP outlet** (s_{OUTLETAP}): Sacrococcygeal joint (X_{cocc}) to inferior pubic symphysis ($X_{\text{pub,inf}}$), with unit direction $\mathbf{e}_{\text{OUTAP}} = (X_{\text{pub,inf}} - X_{\text{cocc}}) / \|X_{\text{pub,inf}} - X_{\text{cocc}}\|$.

For each subject s , patient-specific landmark coordinates follow the registration mapping $X^s = X + \boldsymbol{\varphi}_s(X)$, and the resulting functional observable for a displacement field $\mathbf{u}$ is

$$s_\ell^{(s)}(\mathbf{u}) = \mathbf{e}_\ell^{(s)\top} (\mathbf{u}(X_{\text{target}}^s) - \mathbf{u}(X_{\text{origin}}^s)) = \mathbf{b}_\ell^{(s)\top} \mathbf{u}, \quad \ell \in \{\text{AP, ML, BIS, BIT, OUTLETAP}\}, \quad (18)$$

with measurement vectors $\mathbf{b}_\ell^{(s)} \in \mathbb{R}^n$ constructed using inverse-distance weighting (IDW) interpolation onto the $k_{\text{idw}} = 8$ nearest mesh nodes around each anatomical landmark coordinate, ensuring smooth, mesh-robust landmark displacement estimates across subjects.

Basis-invariant max-nodal anatomical capacity. For any individual functional $\mathbf{b}_\ell^{(s)}$ and an orthonormal basis $Q_s \in \mathbb{R}^{n \times r}$ spanning a clustered subspace of dimension r ($Q_s^\top M_{\text{ref}}^0 Q_s = I_r$), every displacement in the subspace is parameterized by $\mathbf{u} = Q_s \boldsymbol{\alpha}$ with $\boldsymbol{\alpha} \in \mathbb{R}^r$. The maximum anatomical diameter response per 1 mm maximum nodal displacement is defined by the constrained max-nodal capacity:

$$\kappa_{\ell,\infty}(Q_s) = \max_{\boldsymbol{\alpha} \neq \mathbf{0}} \frac{|\mathbf{b}_\ell^{(s)\top} Q_s \boldsymbol{\alpha}|}{\max_a \|(Q_s \boldsymbol{\alpha})_a\|_2} = \max_{\boldsymbol{\alpha}} |\mathbf{b}_\ell^{(s)\top} Q_s \boldsymbol{\alpha}| \quad \text{s.t.} \quad \max_a \|(Q_s \boldsymbol{\alpha})_a\|_2 \leq 1, \quad (19)$$

where a indexes the finite-element mesh nodes and $\|(Q_s \boldsymbol{\alpha})_a\|_2$ denotes the Euclidean displacement norm of node a . Because both the numerator and the max-nodal constraint depend only on the physical displacement field $\mathbf{u} = Q_s \boldsymbol{\alpha}$, this quantity is strictly invariant under arbitrary internal orthogonal basis rotations $\tilde{Q}_s = Q_s R$ with $R \in O(r)$, since $\tilde{Q}_s(R^\top \boldsymbol{\alpha}) = Q_s \boldsymbol{\alpha}$. For $r = 2$, the zero-homogeneity of the objective allows parameterization of $\boldsymbol{\alpha}$ by polar angle $\theta \in [0, \pi)$ with $\boldsymbol{\alpha}(\theta) = (\cos \theta, \sin \theta)^\top$, solved via high-resolution angular search and scalar refinement; for $r \geq 3$, Nelder–Mead simplex optimisation was used to estimate the optimum; multi-start validation on representative subjects agreed to $< 10^{-6}$. The optimal spatial displacement field $\boldsymbol{\psi}_\ell^{(s)} = Q_s \boldsymbol{\alpha}^* / \max_a \|(Q_s \boldsymbol{\alpha}^*)_a\|_2$ satisfies $\max_a \|\boldsymbol{\psi}_{\ell,a}^{(s)}\|_2 = 1.0$ exactly, and the resulting anatomical capacity $\kappa_{\ell,\infty}(Q_s) = |\mathbf{b}_\ell^{(s)\top} \boldsymbol{\psi}_\ell^{(s)}|$ has physical units of millimetres diameter change per millimetre maximum nodal displacement.

Multivariate canonical decomposition. When joint coupling across a pair of measurement vectors $(\mathbf{b}_1^{(s)}, \mathbf{b}_2^{(s)})$ (e.g., AP and ML inlet, or BIS and BIT outlet) is examined, we assemble the covector matrix $B_s = [\mathbf{b}_1^{(s)} \quad \mathbf{b}_2^{(s)}] \in \mathbb{R}^{n \times 2}$ and its subspace projection:

$$C_s = Q_s^\top B_s = [Q_s^\top \mathbf{b}_1^{(s)} \quad Q_s^\top \mathbf{b}_2^{(s)}] \in \mathbb{R}^{r \times 2}, \quad C_s = U_s \Sigma_s V_s^\top, \quad (20)$$

where the singular values $\sigma_1 \geq \sigma_2$ quantify joint extremal capacities. The canonical displacement modes in physical space are $\boldsymbol{\psi}_k^{(s)} = Q_s U_s[:, k]$ ($k = 1, 2$), with corresponding dual functionals in covector space $\tilde{\mathbf{b}}_k^{(s)} = B_s V_s[:, k] = V_s[0, k] \mathbf{b}_1^{(s)} + V_s[1, k] \mathbf{b}_2^{(s)}$. By construction, their action is exactly equal to the singular values: $\tilde{\mathbf{b}}_k^{(s)\top} \boldsymbol{\psi}_k^{(s)} = \sigma_k$. The clean singular scores normalized by maximum nodal displacement are $\hat{s}_k^{(s)} = \sigma_k / \max_a \|\boldsymbol{\psi}_{k,a}^{(s)}\|_2$. Because $V_s \in O(2)$ mixes the two anatomical axes, mode $\boldsymbol{\psi}_1^{(s)}$ is primary-dominant only when $|V_s[0, 0]| \geq |V_s[1, 0]|$; across the pelvic cohort, the first singular mode in the 9–10 subspace is predominantly aligned with the higher-compliance transverse (ML) axis ($|V_s[1, 0]| > |V_s[0, 0]|$ in 99.6% of subjects). For unambiguous, axis-specific evaluation of anatomical opening without coordinate mixing, the basis-invariant single-functional capacity $\kappa_{\ell, \infty}(Q_s)$ (Eq. 19) serves as our primary quantitative metric, while joint-SVD scores provide complementary bivariate structure.

4.9 Whole-pelvis global deformation pattern decomposition

We analysed the fifteen stored reference eigenvectors on the complete reference continuum mesh, without anatomical subdivision. Continuous displacement fields were sampled using positive 14-point tetrahedral quadrature (exact for polynomials up to degree 4; Basix scheme), ensuring exact numerical integration of products between piecewise-linear finite-element fields (P_1) and quadratic polynomial templates (P_2 , degree 3) as well as quadratic-quadratic products ($P_2 \times P_2$, degree 4). Direct projection onto local strain fields $\boldsymbol{\epsilon}_k = \frac{1}{2}(\nabla \mathbf{u}_k + \nabla \mathbf{u}_k^\top)$ leaves 97–99% of the strain norm in the residual across all eigenmodes, because compliance and strain concentrate heavily in the cartilaginous joints (sacroiliac joints and pubic symphysis). This localized joint concentration masks global ring deformation and fails to discriminate modes. To recover the global ring-level character, we evaluate overall shape change across the entire pelvic continuum.

Infinitesimal rigid-body motion (3 translations and 3 rotations) is spanned by the 6-parameter field $\mathbf{u}_{\text{rigid}}(\mathbf{x}) = \mathbf{t} + \boldsymbol{\omega} \times (\mathbf{x} - \mathbf{x}_c)$, where $\mathbf{x}_c = \langle \mathbf{x} \rangle_V$ is the volume-weighted centroid. The volume-weighted rigid design matrix was orthonormalised via reduced QR decomposition ($\mathbf{Q}_{\text{rigid}}$), and rigid motion was projected out from the displacement field:

$$\mathbf{u}_{\text{nonrigid}} = \mathbf{u} - \mathbf{Q}_{\text{rigid}}(\mathbf{Q}_{\text{rigid}}^\top \mathbf{u}). \quad (21)$$

Rigid rotation is therefore completely eliminated upfront and is not one of the reported deformation categories.

Coordinates relative to the centroid $\mathbf{x}_c$ were scaled by the gyration radius $r_g = \sqrt{\langle \|\mathbf{x} - \mathbf{x}_c\|^2 \rangle_V}$, defining normalised coordinates $(\tilde{x}, \tilde{y}, \tilde{z}) = (\mathbf{x} - \mathbf{x}_c)/r_g$ in the anatomical frame (ML, AP, CC). Rather than conditioning on an artificial 1D beam axis, the deformation is decomposed across the entire 3D ring into four canonical deformation families spanning the complete 24-dimensional nonrigid polynomial displacement space up to quadratic order:

- (i) **Axial** (6 DOF): uniform normal extension/compression and normal strain gradients along all three axes, $u_i \in \text{span}\{\tilde{x}_i, \frac{1}{2}\tilde{x}_i^2\}$ for $i \in \{x, y, z\}$;
- (ii) **Bending** (6 DOF): pure Euler-Bernoulli flexural modes with zero shear strain ($\varepsilon_{ij} = 0$ for $i \neq j$), $\mathbf{b}_{ij} = \frac{1}{2}\tilde{x}_i^2 \mathbf{e}_j - \tilde{x}_i \tilde{x}_j \mathbf{e}_i$ for each ordered pair $i \neq j \in \{x, y, z\}$;
- (iii) **Shear** (10 DOF): uniform transverse shears $((\tilde{y}, \tilde{x}, 0), (0, \tilde{z}, \tilde{y}), (\tilde{z}, 0, \tilde{x}))$, six quadratic transverse shear gradients, and the complete irrotational quadratic shear mode $\mathbf{w} = \nabla(\tilde{x}\tilde{y}\tilde{z}) = (\tilde{y}\tilde{z}, \tilde{x}\tilde{z}, \tilde{x}\tilde{y})^\top$ ($d = 10$ total);
- (iv) **Torsion** (2 DOF): Saint-Venant twist gradients with zero normal strain, $\mathbf{t}_x = (0, -\tilde{x}\tilde{z}, \tilde{x}\tilde{y})^\top$ and $\mathbf{t}_y = (\tilde{y}\tilde{z}, 0, -\tilde{x}\tilde{y})^\top$.

Templates in each family were projected orthogonal to $\mathbf{Q}_{\text{rigid}}$ and orthonormalised using reduced singular value decomposition. The combined 24-dimensional nonrigid basis $\mathbf{A}$ spans the deformation space with low condition number ($\kappa(\mathbf{A}) \approx 4.58$ on the reference pelvis), corresponding to a Gram matrix condition number $\kappa(\mathbf{G}) = \kappa(\mathbf{A})^2 \approx 20.98$. Because the four deformation families exhibit intrinsic non-orthogonality (correlated explanatory subspaces) on irregular geometries, Shapley value allocation is employed as an objective game-theoretic method to partition the shared explained norm fairly and symmetrically across the subspaces, rather than acting as a separator of mutually orthogonal physical mechanisms:

$$f_m = \sum_{S \subseteq M \setminus \{m\}} \frac{|S|!(|M| - |S| - 1)!}{|M|!} [R^2(S \cup \{m\}) - R^2(S)], \quad \sum_{m=1}^4 f_m + f_{\text{residual}} = 1, \quad (22)$$

where $R^2(S)$ is the fraction of nonrigid squared norm explained by subspace subset S , and $f_{\text{residual}} = 1 - R^2(M)$. Shapley allocation distributes the explained squared norm without order-dependent orthogonalisation bias. The fractions are invariant to eigenvector sign and amplitude, describe a specified global 3D shape-change model, and are not strain-energy shares. Comprehensive mathematical derivations of the continuum inner product, 14-point tetrahedral quadrature, canonical polynomial dictionary, cooperative game-theoretic Shapley allocation, numerical conditioning, synthetic backward-recovery benchmarks, and complete 15-mode quantitative profiles are detailed in Appendix E (Tables E1–E2).

4.10 Load routing and modal strain-energy fractions

For each load case we solved the static problem $K_s \mathbf{u}_s = \mathbf{b}$ on Ω_{ref} , projected the result onto the M_s -orthonormal eigenbasis Φ ,

$$q = \Phi^\top M_s \mathbf{u}_s, \quad E_i = \frac{1}{2} \lambda_i q_i^2, \quad f_i = \frac{E_i}{\sum_{j=1}^{N_{\text{ev}}} E_j}, \quad (23)$$

and summarised each pathway's contribution by its retained modal strain-energy fraction f_i evaluated across the $N_{\text{ev}} = 15$ computed modes. Block throughput for a clustered regime C is $f_C = \sum_{i \in C} f_i$, and the dominant mode set N_{80} of a load case is the smallest subset of modes (selected in descending order of mean energy contribution) whose cumulative mean energy fractions reach at least 0.80. For cross-load comparison we also report cosine similarity of full 15-mode mean profiles and Jaccard overlap of dominant mode sets.

4.11 First-order eigenvalue sensitivities and analytical uncertainty propagation

To propagate material-parameter uncertainty to the spectrum without repeated finite-element solves, we use classical first-order sensitivities of the generalised eigenproblem (10): for a parameter p_j that affects only the stiffness,

$$\frac{\partial \lambda_m}{\partial p_j} = \frac{\phi_m^\top (\partial K_s / \partial p_j) \phi_m}{\phi_m^\top M_s \phi_m}. \quad (24)$$

For ligament parameters, $\partial K_s / \partial k_{\text{seg}} = \mathbf{e} \mathbf{e}^\top$ and $\partial K_s / \partial T_{\text{seg}} = (I - \mathbf{e} \mathbf{e}^\top) / L$ are aggregated over segments in each bundle using the geometric assembly underlying (6). For cartilage moduli, $\partial K_s / \partial E_{\text{region}}$ is the element-wise elastic stiffness integrated over the corresponding joint domain, scaled by the baseline modulus. For the high-density branch of (4), the chain rule gives $\partial E / \partial \alpha = E / \alpha$ and $\partial E / \partial \beta = E \ln \rho_{\text{ash}}$, integrated over the bone domain. Storing the dimensionless Jacobians

$$J_j^{(m)} = \frac{p_j}{\lambda_m} \frac{\partial \lambda_m}{\partial p_j} \quad (25)$$

once per subject lets us compute first-order log-space variance propagation as $\text{Var}(\log \lambda_m) \approx \mathbf{J}_m^\top \Sigma_{\log p} \mathbf{J}_m$ for any prior covariance $\Sigma_{\log p} = \text{Cov}(\log \mathbf{p})$ on the material parameters, including hierarchical priors that separate global, per-type, and left/right components. For adjacent eigenvalues λ_i, λ_{i+1} , the variance of the gap $\Delta\lambda = \lambda_{i+1} - \lambda_i$ is computed from the combined sensitivity vector $\mathbf{g}_{i,i+1} = \lambda_{i+1} \mathbf{J}_{i+1} - \lambda_i \mathbf{J}_i$ as

$$\text{Var}(\lambda_{i+1} - \lambda_i) \approx \mathbf{g}_{i,i+1}^\top \Sigma_{\log p} \mathbf{g}_{i,i+1} \approx \text{Var}(\lambda_{i+1}) + \text{Var}(\lambda_i) - 2 \text{Cov}(\lambda_i, \lambda_{i+1}), \quad (26)$$

accounting for the substantial positive covariance induced by shared tissue parameters. Fig. 9 summarises the resulting variance decomposition across the 15 lowest eigenvalues and the covariance-aware gap signal-to-noise ratio across solver ranks. For the low-rank backbone (modes 1–3), adjacent gap signal-to-noise ratios remain high ($Z \gg 1.0$), demonstrating that modal ordering is gap-protected against material uncertainty. In contrast, for the 9–10 inlet cluster, the gap signal-to-noise ratio drops below unity ($Z < 1.0$, Fig. 9B), indicating that the nominal spectral gap is comparable to predicted material-parameter uncertainty, consistent with local label exchangeability. Cluster boundaries and subspace definitions remain well-defined under the posited parameter covariances, reinforcing that the compact subspace, rather than the isolated modal label, is the robust physical object. Full hierarchical priors, Shapley-style decompositions, and sex-stratified versions are reported in Appendix B and Appendix D.

4.12 Decomposition of variability sources and robustness analyses

To separate shape and material contributions to spectral variability, we compared the full cohort against a shape-only cohort (χ_s , fixed reference bone modulus) and a material-only cohort (identity deformation, subject-specific bone modulus). Robustness was assessed by (i) random permutation of subject order (invariance to machine precision), (ii) subject bootstrap with replacement ($N=278$, 500 replicates) with threshold re-estimation, alongside random 80% subsampling for cohort stability, and (iii) morphology-axis permutation null tests in which a one-dimensional anatomical proxy μ (e.g. AP inlet diameter) was shuffled 1000 times to assess whether swap-rate localisation along μ exceeded chance (Table A6). Post-hoc allometric regressions of $\log \lambda_k$ on isotropic body-size scale s (with instrumental variables from volume, surface, and mass) and Shapley/LMG variance decomposition into scale, sex, shape, and material contributions are reported in Appendix D.

4.13 Minimal analytical model for the near-degenerate regime

The cluster/subspace framework admits a compact analytical interpretation via a two-degree-of-freedom model. Let q_1, q_2 denote two generalised rotational coordinates and define a stiffness matrix

$$K_2 = \begin{bmatrix} k_1 & c \\ c & k_2 \end{bmatrix}, \quad \Delta = k_1 - k_2, \quad k_1, k_2 > 0, \quad k_1 k_2 > c^2, \quad (27)$$

with detuning Δ and coupling c . The eigenvalues are

$$\lambda_{\pm} = \frac{k_1 + k_2}{2} \pm \sqrt{\left(\frac{\Delta}{2}\right)^2 + c^2}, \quad \tan(2\theta) = \frac{2c}{\Delta}, \quad (28)$$

and the minimum separation $\lambda_+ - \lambda_- = 2|c|$ is attained at $\Delta = 0$, where the mixing angle passes through $\pi/4$ and the modes exchange character. The operational criterion

$$|\Delta(\mu)| \leq \tau |c(\mu)| \quad \text{for some morphology parameter } \mu, \quad \tau \in [1.5, 2], \quad (29)$$

separates near-degenerate/mixing from stable-ordering regimes. Adding a third coordinate q_3 with stiffness s and coupling vector $\mathbf{d} = [d_1, d_2]^\top$, dynamic reduction of the 3×3 eigenproblem

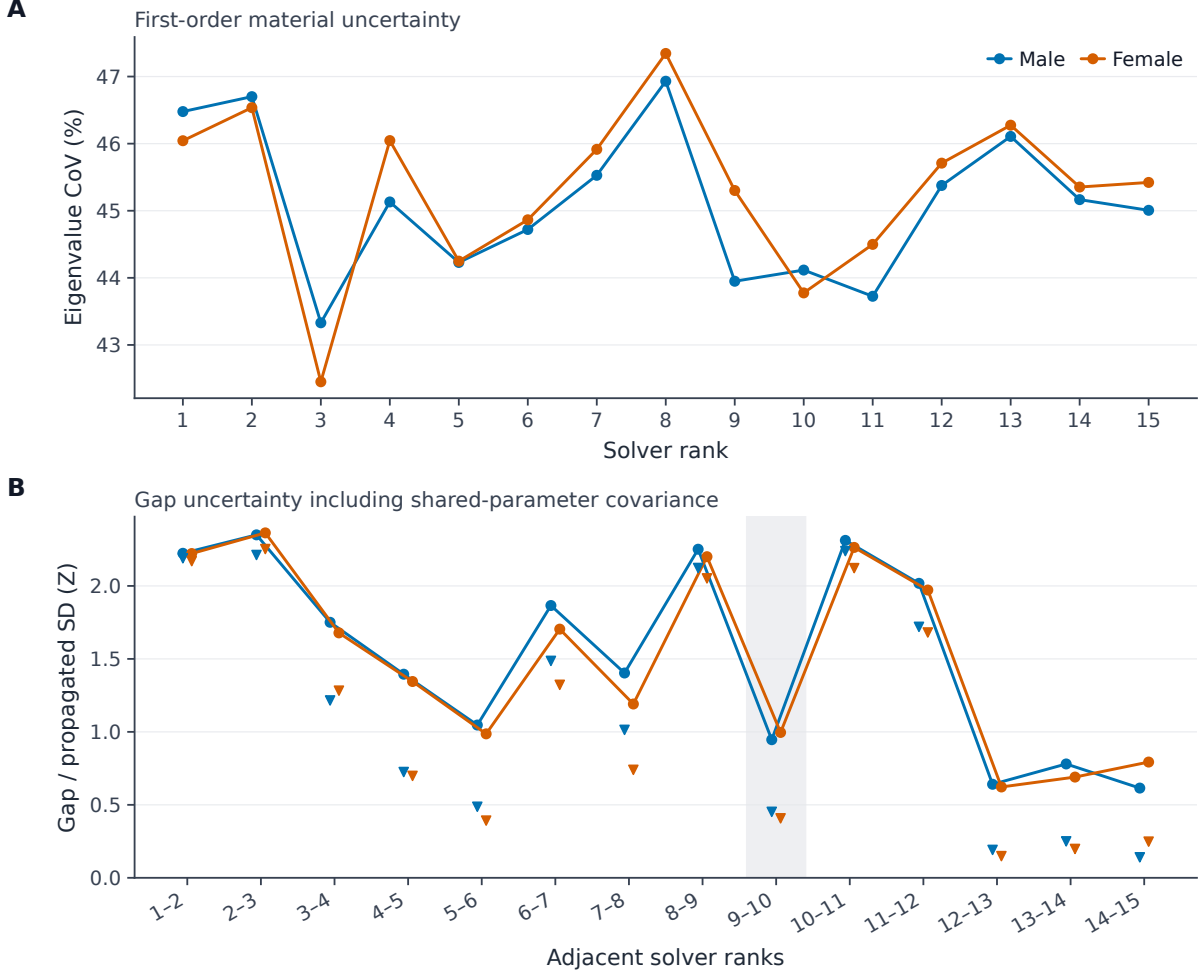


Figure 9: **Analytical propagation of material-parameter uncertainty to eigenvalues and spectral gaps.** (A) Total coefficient of variation (CoV, %) of eigenvalues $\lambda_1, \dots, \lambda_{15}$ under hierarchical material-parameter covariance, stratified by sex (blue: male; yellow: female). (B) Relative gap signal-to-noise ratio (spectral gap divided by propagated standard deviation) between adjacent solver ranks, incorporating shared-parameter covariance $\text{Cov}(\lambda_i, \lambda_{i+1})$. The shaded region highlights the 9–10 rank transition. Triangles mark 10th-percentile values across subjects.

$(K - \lambda I)\mathbf{q} = \mathbf{0}$ gives the exact non-linear reduced eigenproblem

$$\left(K_2 - \frac{\mathbf{d}\mathbf{d}^\top}{s - \lambda}\right)\mathbf{q}_{12} = \lambda\mathbf{q}_{12}. \quad (30)$$

In the low-frequency limit ($\lambda \ll s$), static condensation yields the linear effective stiffness

$$K_{\text{eff}} = K_2 - \frac{\mathbf{d}\mathbf{d}^\top}{s} = \begin{bmatrix} k_1 - d_1^2/s & c - d_1 d_2/s \\ c - d_1 d_2/s & k_2 - d_2^2/s \end{bmatrix}, \quad (31)$$

showing how a compliant hidden pathway (small s) shifts effective detuning and coupling and pushes a system toward an avoided crossing veering region. In all stochastic numerical simulations (Fig. 7), the full 3×3 eigenproblem is solved directly without static or dynamic truncation. In the n -DOF finite-element setting, the detected clusters correspond to $\{q_1, q_2\}$, the canonical basis (ψ_1, ψ_2) to the rotated eigenvectors, and gap_{in} to $(\lambda_+ - \lambda_-)/\lambda_-$. A stochastic population instantiation of this model, shown and discussed in Fig. 7 and Sec. 2.7, reproduces the label-subspace dissociation, the standing-vs-task routing contrast, and the effect of hidden-pathway

variability observed in the full pelvic cohort. Explicit derivations, the small- c expansion, and the full stochastic population extension are provided in Appendix D.

4.14 Statistical analysis

All statistical analyses were executed using SciPy, statsmodels, and custom Python routines. Continuous metrics across non-normal cohorts were evaluated using non-parametric methods. Two-group comparisons (e.g., non-swapped vs. swapped cluster members, or male vs. female distributions) were conducted using two-sided Mann–Whitney U tests. Effect sizes for Mann–Whitney tests are reported as rank-biserial correlations, $r_{\text{rb}} = 1 - 2U/(n_1 n_2) \in [-1, 1]$, and Cliff’s δ . Paired differences across loading conditions or mode pairs were assessed via two-sided Wilcoxon signed-rank tests. Bivariate associations between continuous anatomical dimensions (such as AP inlet diameter or isotropic scale) and modal metrics (energy fractions, spectral gaps) were quantified using Spearman’s rank correlation coefficient ρ_s . Multiple hypothesis testing across spectral ranks and loading cases was controlled using the Benjamini–Hochberg procedure, maintaining the false discovery rate at $q < 0.05$; adjusted p -values are reported where indicated. Confidence intervals for median differences and effect sizes were obtained via bias-corrected and accelerated (BCa) bootstrap resampling (10 000 resamples). Cohort-level cluster stability was evaluated by repeating the detection pipeline across 500 random subcohorts of 80% sample size drawn without replacement ($n = 222$), yielding 95% subsampling stability intervals. Full population inference was additionally confirmed by resampling subjects with replacement ($N = 278$, 500 replicates) with end-to-end recalculation of the cluster threshold ε_{in} .

Differences in subspace orientation across the cohort were quantified using the Grassmann distance $d_G(Q_{\text{ref}}, Q_s) = \|\theta\|_2$. Importantly, d_G measures the intrinsic geometric distance between the linear subspaces $\text{span}(Q_{\text{ref}})$ and $\text{span}(Q_s)$ in the Grassmannian manifold $\text{Gr}(r, n)$, which is strictly invariant to arbitrary unitary rotations of the internal basis within either subspace. Consequently, population variations in d_G (such as sex-specific differences) represent true rotations of the subspace orientation within $\mathbb{R}^n$, rather than artefacts of internal basis re-indexing.

4.15 Software and computational implementation

Finite-element assembly, eigensolves, and postprocessing were implemented in Python 3.11 with DOLFINx/FEniCSx (≥ 0.10), PETSc, and SLEPc. All reported figures, tables, and numerical values are produced by scripts included in the study repository. Results are stored as compressed Zarr arrays for reproducible downstream analysis. Further implementation details, dependency versions, and reproduction instructions are provided in the appendices and the code repository.

Data availability

Derived summary data supporting the findings of this study are provided within the manuscript, the appendices, and the study repository. The standardised pelvic geometries originate from the BoneDat research dataset [3]; access to the underlying clinical imaging data remains subject to source-data governance restrictions and requires approval through the procedures described in the dataset publication.

Code availability

Custom code used to assemble the models, run the spectral analyses, and generate the reported figures and tables is available in the study repository. A public archival release of the analysis code will accompany publication.

Author contributions

P.H., M.K., and N.H. conceptualised the study. P.H. and M.K. developed the computational models and performed the simulations. P.H., M.K., and A.M. analysed the data. P.H., M.K., A.M., and N.H. interpreted the results and wrote the manuscript. All authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Acknowledgments

We thank the BoneDat project for curating the standardised imaging-derived dataset used in this study.

A Spectral cluster architecture

The Supporting Information appendices collect detailed robustness, specificity, and model-boundary analyses that support the findings presented in the main text.

This appendix expands the bookkeeping behind the recurrent clustered regimes, including reference-mode occupancy, swap frequencies, and morphology-axis diagnostics. Table A1 reproduces the cohort-level cluster list used in the main Results. Table A2 records the mapping between recurrent rank clusters and their dominant reference-mode identities. Table A3 reports per-mode swap rates, Table A4 gives the adjacent rank-pair diagnostics, Table A5 summarises the exchange-regime classification, and Table A6 reports the morphology-axis permutation null test used to localise swap behaviour.

Table A1. Detected near-degenerate rank clusters

Cluster	Dim	n	Prev.	Med. gap _{in,min}	Med. sep _{out}	Type
Rank modes 9–10	2	152	54.7%	0.08114	0.4926	near-degenerate
Rank modes 12–15	4	116	41.7%	0.03153	NaN	near-degenerate
Rank modes 5–6	2	94	33.8%	0.08737	0.2345	near-degenerate
Rank modes 12–13	2	78	28.1%	0.03839	0.1941	near-degenerate
Rank modes 14–15	2	70	25.2%	0.05116	NaN	near-degenerate
Rank modes 12–14	3	45	16.2%	0.04109	0.1855	near-degenerate
Rank modes 4–5	2	35	12.6%	0.09521	0.2185	near-degenerate
Rank modes 13–15	3	29	10.4%	0.04465	NaN	near-degenerate
Rank modes 7–8	2	21	7.6%	0.1051	0.3676	near-degenerate

Table A2. Rank-cluster to reference-mode mapping

rank_cluster	dim	n_subjects	prevalence	paired_all_rate	ref_modes_mode	ref_modes_mode_rate	ref_modes_top3
Rank modes 9–10	2	152	54.7%	1	9–10	1	9–10 (100%)
Rank modes 12–15	4	116	41.7%	0.7414	12–15	1	12–15 (100%)
Rank modes 5–6	2	94	33.8%	1	5–6	0.8298	5–6 (83%); 4,6 (17%)
Rank modes 12–13	2	78	28.1%	0.9872	13–14	0.4805	13–14 (48%); 12–13 (44%); 12,15 (6%)
Rank modes 14–15	2	70	25.2%	0.8571	12,15	0.45	12,15 (45%); 14–15 (38%); 13–14 (15%)
Rank modes 12–14	3	45	16.2%	0.9333	12–14	0.8571	12–14 (86%); 12,13,15 (10%); 13–15 (5%)
Rank modes 4–5	2	35	12.6%	1	4–5	1	4–5 (100%)
Rank modes 13–15	3	29	10.4%	0.8966	12–14	0.5	12–14 (50%); 13–15 (31%); 12,14,15 (15%)
Rank modes 7–8	2	21	7.6%	1	7–8	1	7–8 (100%)

Table A3. Per-mode swap rates

Mode	Swap rate	Fail rate	Non-id. rate	n swapped	n failed
1	0	0	0	0	0
2	0	0	0	0	0
3	0.003597	0	0.003597	1	0
4	0.2698	0	0.2698	75	0
5	0.3777	0	0.3777	105	0
6	0.1835	0	0.1835	51	0
7	0.01079	0	0.01079	3	0
8	0.003597	0	0.003597	1	0
9	0.2698	0	0.2698	75	0
10	0.2698	0	0.2698	75	0
11	0.003597	0	0.003597	1	0
12	0.6151	0.007194	0.6223	171	2
13	0.6295	0	0.6295	175	0
14	0.5072	0.01439	0.5216	141	4
15	0.3813	0.1978	0.5791	106	55

Table A4. Adjacent rank-pair swap diagnostics

cluster	cluster_key	rank_pair	ref_pair_mode	swap_rate	invalid_rate	match_rate	median_gap_in	near_deg_rate	true_deg_rate	peak_rate	active_frac	cv_swap	verdict
Rank modes 9–10	modes_9_10	9 $\rightleftharpoons$ 10	9 $\rightleftharpoons$ 10	0.3750	0.0000	1.0000	0.1206	0.5468	0.0000	1.0000	0.2050	0.6157	veering
Rank modes 12–15	modes_12_15	12 $\rightleftharpoons$ 13	12 $\rightleftharpoons$ 13	0.4154	0.0345	0.5603	0.0516	0.8597	0.0000	1.0000	0.3094	0.7588	veering
Rank modes 12–15	modes_12_15	13 $\rightleftharpoons$ 14	13 $\rightleftharpoons$ 14	0.1429	0.0603	0.3017	0.0802	0.7014	0.0000	1.0000	0.0324	2.0521	veering
Rank modes 12–15	modes_12_15	14 $\rightleftharpoons$ 15	14 $\rightleftharpoons$ 15	0.1818	0.2241	0.3793	0.0721	0.7734	0.0000	1.0000	0.0935	1.6757	veering
Rank modes 5–6	modes_5_6	5 $\rightleftharpoons$ 6	5 $\rightleftharpoons$ 6	0.1282	0.0000	0.8298	0.1661	0.3633	0.0000	1.0000	0.0288	1.6080	veering
Rank modes 12–13	modes_12_13	12 $\rightleftharpoons$ 13	13 $\rightleftharpoons$ 14	0.0811	0.0128	0.4744	0.0516	0.8597	0.0000	1.0000	0.0432	2.9996	veering
Rank modes 14–15	modes_14_15	14 $\rightleftharpoons$ 15	12 $\rightleftharpoons$ 15	0.6296	0.1429	0.3857	0.0721	0.7734	0.0000	1.0000	0.3921	0.8933	veering

Table A5. Cluster-level exchange classification

cluster	cluster_key	dim	n_subjects	prevalence	median_gap_in_min	gap_type	swap_verdict	find_verdict	ref_positives_posn	ref_positives_posn_rate	paired_d_rate	near_posn_rate	subd_rate	within_block_rate	transposition_rate	cycle_2_rate	n_pair_posn	n_veering_rate
Rank modes 9–10	modes_9_10	2	152	54.7%	0.0811	non-degenerate	VEERING (localised character exchange)	VEERING (localised character exchange)	9–10	1.0000	1.0000	0.3750	0.0000	1.0000	0.3750	0.0000	1	1
Rank modes 12–15	modes_12_15	4	116	41.7%	0.0315	non-degenerate	VEERING (localised character exchange)	MIXING (multi-mode cycles ; 2 common)	12–15	1.0000	0.7414	0.5431	0.2586	0.7414	0.3023	0.3372	3	3
Rank modes 5–6	modes_5_6	2	94	31.3%	0.0874	non-degenerate	VEERING (localised character exchange)	VEERING (localised character exchange)	5–6	0.8298	1.0000	0.1661	0.0000	0.8298	0.1282	0.0000	1	1
Rank modes 12–13	modes_12_13	2	78	28.1%	0.0384	non-degenerate	VEERING (localised character exchange)	VEERING (localised character exchange)	13–14	0.4805	0.9872	0.0385	0.0128	0.4744	0.0811	0.0000	1	1
Rank modes 14–15	modes_14_15	2	70	25.2%	0.0512	non-degenerate	VEERING (localised character exchange)	VEERING (localised character exchange)	12–15	0.4509	0.8571	0.1429	0.1429	0.3857	0.3704	0.0000	1	1

Table A6. Morphology-axis permutation null test

Cluster	Rank pair	Peak _{obs}	<i>p</i> (peak)	Active _{obs}	<i>p</i> (active)	CV _{obs}	<i>p</i> (CV)	Significant
Rank modes 9–10	9 $\rightleftharpoons$ 10	1	0.2547	0.2162	0.8202	0.6042	0.02697	NO
Rank modes 12–15	12 $\rightleftharpoons$ 13	1	0.4496	0.283	0.7223	0.5553	0.4046	NO
Rank modes 12–15	13 $\rightleftharpoons$ 14	0.25	0.8891	0.5	0.3437	1	0.4336	NO
Rank modes 12–15	14 $\rightleftharpoons$ 15	0.5	0.7443	0.09302	0.8492	0.9727	0.4256	NO
Rank modes 5–6	5 $\rightleftharpoons$ 6	0.5	0.9001	0.1267	0.4535	1.048	0.7263	NO
Rank modes 12–13	12 $\rightleftharpoons$ 13	0.25	0.8192	0.4375	0.2438	1.146	0.5455	NO
Rank modes 14–15	14 $\rightleftharpoons$ 15	0.75	0.6563	1	0.5504	0	1	NO

B Cohort robustness and sex effects

The main text uses sex effects only where they materially sharpen interpretation of the recurrent clustered subspaces, especially the 9–10 inlet-related pair. This appendix provides the broader sex-stratified summary and the subsampling stability evidence that the dominant clusters are stable cohort-level features rather than products of a few outlying subjects. Table B1 gives the subsampling stability estimates for the dominant clusters and gap statistics. Figure B1 summarises the sex-stratified spectral structure, and Table B2 reports the accompanying sex-stratified statistical tests for the spectral and subspace metrics.

Table B1. Subsampling stability under random 80% subcohorts

Metric	Target	Observed	Boot. mean	Boot. std	95
prevalence	Rank modes 9–10	0.547	0.545	0.035	[0.482, 0.617]
prevalence	Rank modes 12–15	0.417	0.417	0.033	[0.356, 0.486]
prevalence	Rank modes 5–6	0.338	0.337	0.031	[0.279, 0.401]
prevalence	Rank modes 12–13	0.281	0.280	0.029	[0.225, 0.340]
prevalence	Rank modes 14–15	0.252	0.252	0.028	[0.198, 0.311]
swap_rate	Mode 4	0.270	0.271	0.030	[0.216, 0.329]
swap_rate	Mode 5	0.378	0.378	0.033	[0.315, 0.444]
swap_rate	Mode 6	0.183	0.182	0.026	[0.135, 0.239]
swap_rate	Mode 7	0.011	0.011	0.007	[0.000, 0.027]
swap_rate	Mode 9	0.270	0.271	0.028	[0.218, 0.331]
swap_rate	Mode 10	0.270	0.271	0.028	[0.218, 0.331]
swap_rate	Mode 12	0.594	0.595	0.033	[0.523, 0.658]
swap_rate	Mode 13	0.622	0.622	0.033	[0.554, 0.689]
swap_rate	Mode 14	0.468	0.470	0.033	[0.405, 0.545]
swap_rate	Mode 15	0.453	0.455	0.035	[0.390, 0.523]
median_gap	Pair 1–2	1.609	1.603	0.038	[1.521, 1.665]
median_gap	Pair 2–3	1.183	1.189	0.019	[1.159, 1.221]
median_gap	Pair 3–4	0.484	0.483	0.020	[0.448, 0.526]
median_gap	Pair 4–5	0.271	0.271	0.011	[0.247, 0.295]
median_gap	Pair 5–6	0.166	0.166	0.008	[0.148, 0.183]
median_gap	Pair 6–7	0.526	0.527	0.013	[0.502, 0.556]
median_gap	Pair 7–8	0.287	0.287	0.011	[0.268, 0.313]
median_gap	Pair 8–9	0.913	0.911	0.014	[0.882, 0.938]
median_gap	Pair 9–10	0.121	0.121	0.007	[0.109, 0.135]
median_gap	Pair 10–11	1.144	1.147	0.021	[1.109, 1.189]
median_gap	Pair 11–12	0.705	0.705	0.014	[0.672, 0.732]
median_gap	Pair 12–13	0.052	0.052	0.004	[0.046, 0.060]
median_gap	Pair 13–14	0.080	0.080	0.005	[0.072, 0.091]
median_gap	Pair 14–15	0.072	0.071	0.006	[0.062, 0.082]

Figure B1. Sex-stratified spectral summary

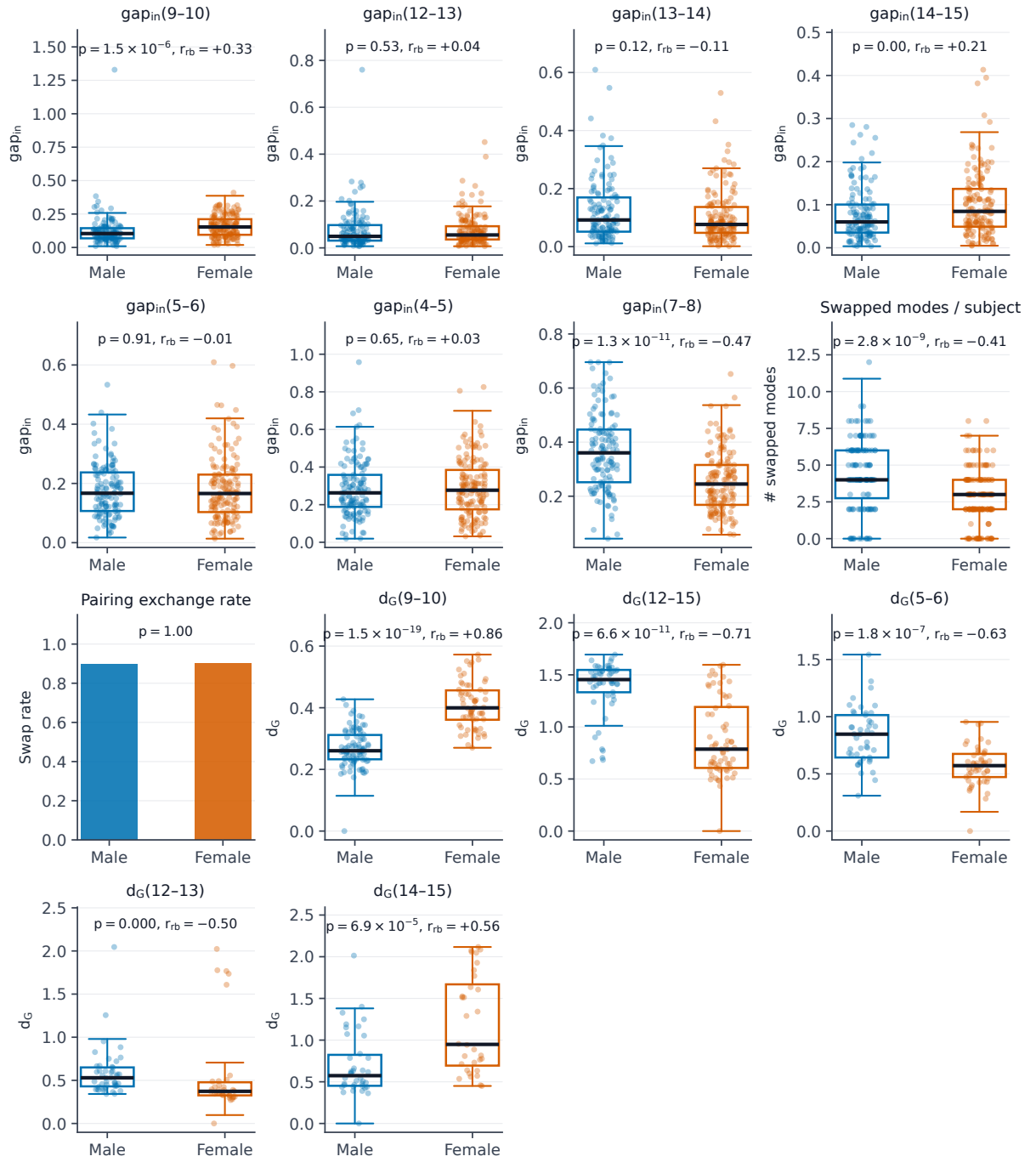


Table B2. Sex-stratified tests for spectral and subspace metrics

Metric	Male med.	Female med.	U	p	r_{rb}
gap_in (9–10)	0.104	0.1534	6384	1.489e-06	0.335
gap_in (12–13)	0.04925	0.0552	9182	0.5321	0.04354
gap_in (13–14)	0.09191	0.07643	1.063e+04	0.1233	-0.1073
gap_in (14–15)	0.06002	0.08448	7631	0.003216	0.2051
gap_in (5–6)	0.1669	0.166	9677	0.9088	-0.008021
gap_in (4–5)	0.2627	0.2768	9299	0.6529	0.03135
gap_in (7–8)	0.3604	0.245	1.412e+04	1.311e-11	-0.471
swap_count_per_subject	4	3	1.352e+04	2.824e-09	-0.4079
permutation_rate	0.8984	0.9	0.9829	1	-0.01724
d_G (modes_9_10)	0.2605	0.3996	377	1.5e-19	0.8649
d_G (modes_12_15)	1.455	0.7866	2848	6.629e-11	-0.7059
d_G (modes_5_6)	0.8472	0.572	1793	1.766e-07	-0.6263
d_G (modes_12_13)	0.5299	0.3729	1120	0.0001815	-0.4973
d_G (modes_14_15)	0.5736	0.9485	270	6.922e-05	0.5559

C Load-routing specificity

The main manuscript uses standing and proxy childbirth loads to show the contrast between a robust low-rank backbone and selectively recruited higher-rank subspaces. This appendix provides comprehensive load-specificity diagnostics, including cluster-level function-locking, dominant mode sets, and pairwise overlap of full routing profiles.

Figure C1 visualises the load–function locking map across the prevalent clusters. Table C1 summarises the phase-specific coupling synthesis, Table C2 reproduces the dominant mode-set summary used in the main text for appendix-level completeness, and Table C3 reports the pairwise overlap metrics for the full load-routing profiles.

Figure C1. Load-function coupling heatmap across prevalent clusters

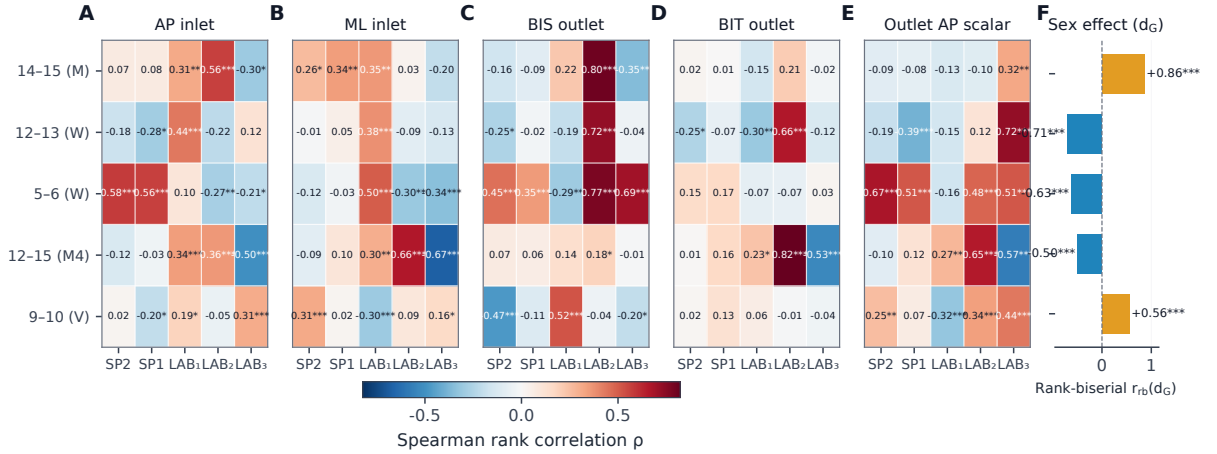


Table C1. Load-function coupling synthesis

Cluster	Regime	Best-predictor coupling axis for FLI (AP/ML inlet, BIS/BIT outlet, or Outlet AP scalar)	FLI ρ	Load case	$\bar{f}_C$	$r_{rs}(d_G)$
Rank modes 9–10	Veering	BIS outlet	0.5153	LAB ₁	0.5291	0.8649
Rank modes 9–10	Veering	Outlet AP scalar	0.339	LAB ₂	0.1578	0.8649
Rank modes 9–10	Veering	Outlet AP scalar	0.4354	LAB ₃	0.1407	0.8649
Rank modes 12–15	Mixing (4D)	AP inlet	0.3442	LAB ₁	0.07568	-0.7059
Rank modes 12–15	Mixing (4D)	BIT outlet	0.8222	LAB ₂	0.01016	-0.7059
Rank modes 12–15	Mixing (4D)	ML inlet	-0.6724	LAB ₃	0.06857	-0.7059
Rank modes 5–6	Weak exchange	ML inlet	0.4989	LAB ₁	0.02502	-0.6263
Rank modes 5–6	Weak exchange	BIS outlet	0.7738	LAB ₂	0.5917	-0.6263
Rank modes 5–6	Weak exchange	BIS outlet	0.6931	LAB ₃	0.217	-0.6263
Rank modes 12–13	Weak exchange	AP inlet	0.4367	LAB ₁	0.06863	-0.4973
Rank modes 12–13	Weak exchange	BIS outlet	0.7197	LAB ₂	0.0009939	-0.4973
Rank modes 12–13	Weak exchange	Outlet AP scalar	0.7232	LAB ₃	0.01162	-0.4973
Rank modes 14–15	Mixing	ML inlet	0.3535	LAB ₁	0.02817	0.5559
Rank modes 14–15	Mixing	BIS outlet	0.8028	LAB ₂	0.01237	0.5559
Rank modes 14–15	Mixing	BIS outlet	-0.3534	LAB ₃	0.008386	0.5559

Table C2. Dominant mode sets for standing and proxy loading

Load	Top mode	Top mode share (%)	Top-3 modes	Top-3 share (%)	N_{80}	Modes in 80% set	Cumulative share at N_{80} (%)
SP2leg	2 (ML rotation)	84.49	2,4,5	93.79	1	2	84.49
SP1leg	1 (CC rotation)	39.89	1,2,3	92.04	3	1,2,3	92.04
LAB ₁	9 (inlet opening)	33.49	9,10,8	66.12	5	9,10,8,4,3	81.48
LAB ₂	5	25.15	5,6,4	74.77	4	5,6,4,9	85.8
LAB ₃	2 (ML rotation)	45.81	2,5,4	68.09	5	2,5,4,9,6	85.8

Table C3. Pairwise overlap of load-routing profiles

Load A	Load B	Jaccard overlap (80% sets)	Cosine similarity (full profile)	Shared modes (80% sets)
SP2leg	SP1leg	0.3333	0.6668	2
SP2leg	LAB ₁	0	0.1202	
SP2leg	LAB ₂	0	0.1343	
SP2leg	LAB ₃	0.2	0.9367	2
SP1leg	LAB ₁	0.1429	0.1233	3
SP1leg	LAB ₂	0	0.1051	
SP1leg	LAB ₃	0.1429	0.6326	2
LAB ₁	LAB ₂	0.2857	0.48	4,9
LAB ₁	LAB ₃	0.25	0.3613	4,9
LAB ₂	LAB ₃	0.8	0.4456	4,5,6,9

D Model-boundary and sensitivity analyses

These analyses bound the main claims rather than extend them. Allometry quantifies what is lost when the main spectral analysis is performed on size-normalised pelves. Material uncertainty quantifies how much absolute eigenvalue identifiability would degrade under broader soft-tissue and density-law scatter while showing that the relative fragile-versus-robust gap architecture remains robust. Figure D1 shows the post-hoc allometric recovery analysis, and Table D1 summarises the corresponding scaling exponents and variance decomposition. Figure D2 presents the threshold sensitivity, reference sensitivity, isotropic scale vs solver gap, and 15-mode displacement reconstruction error, alongside Table D2 which tabulates the combined first-order uncertainty propagation used to support the robustness discussion in the main text.

Figure D1. Allometry dashboard

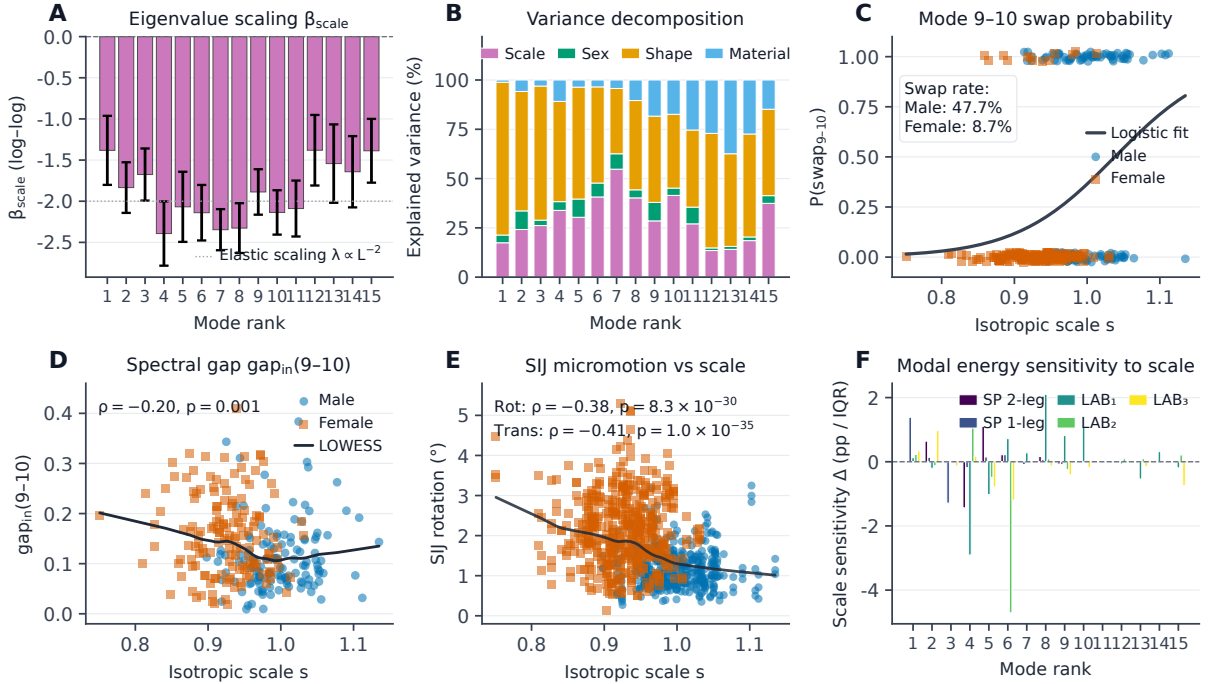


Table D1. Allometry summary

Table 3: Allometric scaling exponents and variance decomposition per eigenvalue mode. β_{scale} is the IV-2SLS log-log coefficient of eigenvalue on isotropic scale, with HC-robust 95 % CIs and FDR-corrected p -values. $\Delta\lambda$ per IQR gives the percent eigenvalue change for an interquartile range shift in log-scale. R^2 columns show the LMG (Shapley) variance share for each predictor group.

Mode	β_{scale}	95% CI	p_{FDR}	$\Delta\lambda$ IQR (%)	R^2 scale (%)	R^2 shape (%)	R^2 material (%)	R^2 sex (%)
1	-1.38	[-1.80, -0.96]	4.7e-10	-10.1	17.4	77.5	1.1	3.9
2	-1.83	[-2.14, -1.53]	8.4e-26	-13.1	24.2	60.6	5.8	9.4
3	-1.68	[-1.99, -1.36]	2.0e-21	-12.1	26.2	68.0	3.1	2.7
4	-2.39	[-2.78, -2.00]	7.7e-27	-16.8	33.9	50.8	10.9	4.4
5	-2.07	[-2.49, -1.64]	1.1e-18	-14.7	30.4	56.8	3.6	9.3
6	-2.14	[-2.48, -1.80]	3.4e-28	-15.1	40.7	48.8	3.5	7.1
7	-2.35	[-2.60, -2.10]	5.7e-49	-16.5	54.8	33.1	4.3	7.8
8	-2.33	[-2.63, -2.02]	2.4e-37	-16.4	40.2	45.3	10.4	4.1
9	-1.89	[-2.16, -1.61]	1.9e-31	-13.5	28.5	43.8	18.3	9.4
10	-2.14	[-2.41, -1.87]	7.7e-39	-15.1	41.5	37.4	17.4	3.7
11	-2.09	[-2.43, -1.75]	6.3e-27	-14.8	27.0	39.1	25.4	8.5
12	-1.38	[-1.81, -0.95]	9.4e-10	-10.1	13.5	58.1	27.1	1.3
13	-1.54	[-2.02, -1.07]	8.2e-10	-11.2	14.1	47.0	37.4	1.5
14	-1.64	[-2.07, -1.21]	1.5e-12	-11.8	18.5	52.2	27.4	1.9
15	-1.39	[-1.77, -1.00]	1.8e-11	-10.1	37.4	43.8	14.8	3.9

Figure D2. Robustness and reconstruction dashboard

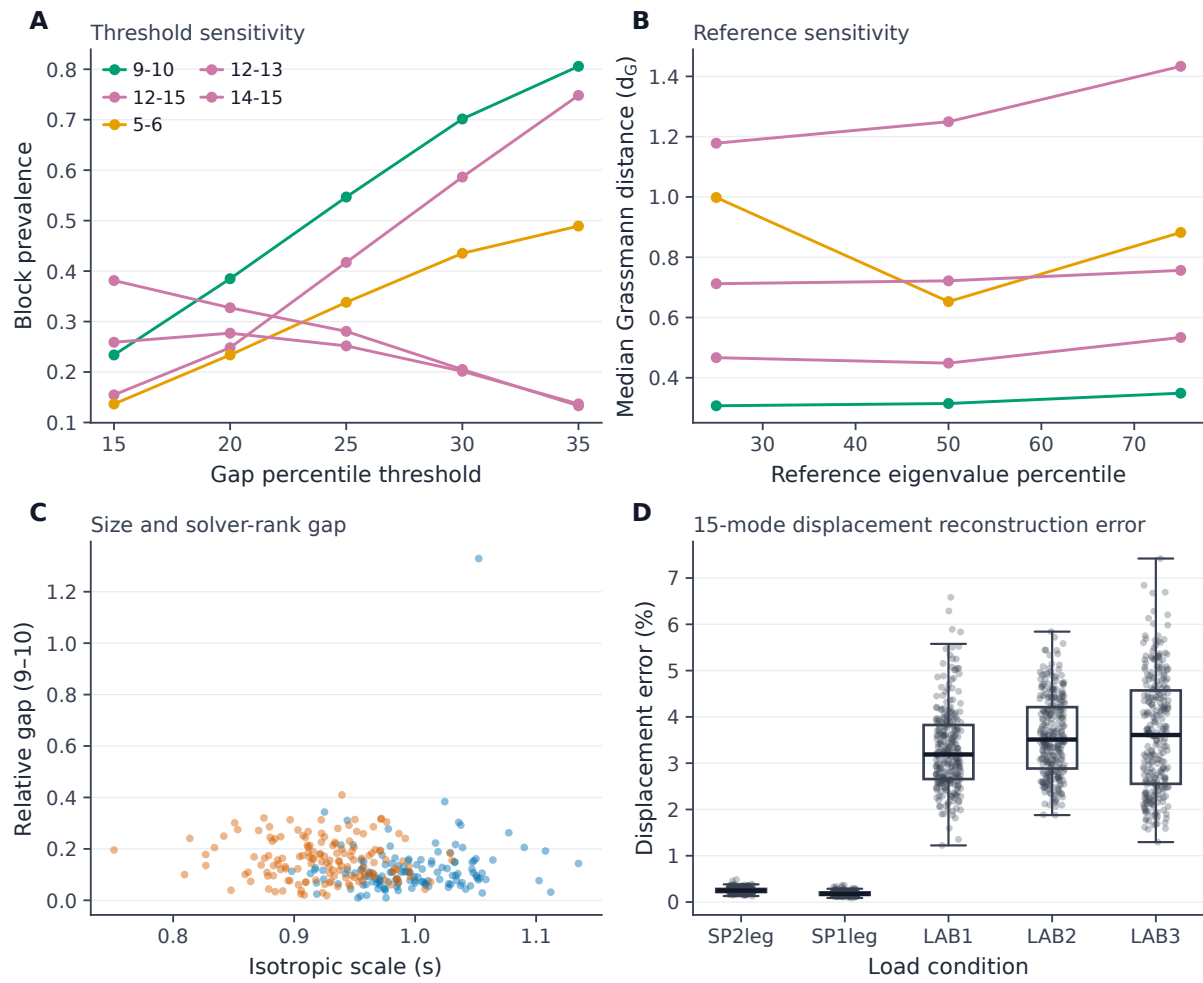


Table D2. Combined first-order material uncertainty summary

Table 4: Population-level material uncertainty propagation stratified by sex. CoV is the total coefficient of variation (ligament + cartilage + bone). Lig/Cart/Bone columns show each group’s fraction of total variance. Gap Z -score measures robustness of each spectral gap to combined material noise (first-order signal-to-noise ratio; not a confidence probability).

Mode	Female ($N = 150$)				Male ($N = 128$)			
	CoV (%)	Lig (%)	Cart (%)	Bone (%)	CoV (%)	Lig (%)	Cart (%)	Bone (%)
1	46.0	5	94	0	46.5	7	93	1
2	46.5	4	93	2	46.7	5	93	2
3	42.5	5	94	1	43.3	6	93	1
4	46.0	6	90	4	45.1	8	90	2
5	44.2	8	89	3	44.2	9	89	2
6	44.9	9	89	2	44.7	10	88	2
7	45.9	16	83	1	45.5	15	84	1
8	47.3	7	90	3	46.9	6	91	3
9	45.3	4	88	7	43.9	4	90	5
10	43.8	2	90	8	44.1	5	90	6
11	44.5	1	87	12	43.7	3	89	9
12	45.7	1	85	14	45.4	2	85	13
13	46.3	1	84	15	46.1	1	83	16
14	45.4	1	86	13	45.2	2	87	10
15	45.4	2	88	10	45.0	3	89	8

E Whole-pelvis 3D canonical deformation pattern decomposition

This appendix provides the complete mathematical formulation, continuum quadrature proofs, coordinate transformations, canonical polynomial dictionary, cooperative game-theoretic projection mechanics, numerical conditioning diagnostics, and synthetic backwards-identification benchmarks for the whole-pelvis 3D deformation decomposition presented in Section 2.1 and Figure 1. Table E1 tabulates the complete 15-mode quantitative profile across all four canonical deformation families, standalone R^2 values, and residual fractions. Table E2 reports the synthetic benchmark validations across idealized symmetric and irregular pelvic domains.

E.1 Physical motivation: Cartilage compliance concentration vs. ring-level kinematics

The pelvic ring is a closed, doubly-curved structural biomechanical loop composed of three major bony segments (the sacrum and bilateral coxal bones) joined anteriorly at the pubic symphysis and posteriorly at the bilateral sacroiliac joints (SIJs). In continuum mechanics, the standard approach to characterise modal kinematics would be to evaluate the infinitesimal Cauchy strain tensor field,

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^\top). \quad (32)$$

However, while physiological dynamic elastic moduli of articular cartilage and symphyseal fibrocartilage are frequently reported in the range $E_{\text{cartilage}} \sim 10\text{--}50$ MPa, in our quasi-static equilibrium finite-element formulations an effective relaxed equilibrium modulus of $E_{\text{cartilage}} = 1.0$ MPa is prescribed for the sacroiliac cartilage cleft and pubic symphysis disc. This effective modulus accounts for sustained equilibrium compliance across the pelvic syndesmoses, which are three to four orders of magnitude more compliant than cortical and trabecular bone ($E_{\text{bone}} \sim 5\text{--}18$ GPa). Under generalised modal vibration, elastic deformation naturally channels into these narrow compliant interfaces. Consequently, direct L^2 -projection of the Cauchy strain field onto low-order polynomial strain templates leaves between 97% and 99% of the strain norm unexplained in the residual across all fifteen computed eigenmodes. The intense local strain spikes in the millimeter-thin joint clefts completely overpower and conceal the systemic, ring-scale kinematics (such as whole-pelvis flaring, torsion, and inlet/outlet dilation).

Furthermore, modeling the human pelvis as a one-dimensional beam conditioned along an arbitrary anatomical axis (e.g., mediolateral, anteroposterior, or craniocaudal) imposes an unphysical 1D kinematic constraint on a closed, three-dimensional ring. For complex three-dimensional deformation pathways involving coupled out-of-plane flexure and transverse shearing (such as Mode 2), 1D axis conditioning artificially drives the unexplained residual up to 84%, distorting biological interpretation. To overcome both limitations, our framework evaluates the overall shape change across the entire three-dimensional pelvic continuum without anatomical partitioning and without imposing artificial 1D beam axes, removing infinitesimal rigid-body motion upfront before decomposing the remaining shape change into four canonical 3D polynomial deformation families.

E.2 Continuum inner product and 14-point tetrahedral quadrature

Let $\Omega \subset \mathbb{R}^3$ denote the reference pelvic continuum domain with volume $V = \int_{\Omega} dV$. The natural inner product on the displacement space $[L^2(\Omega)]^3$ weighted by volume is

$$\langle \mathbf{u}, \mathbf{v} \rangle_V = \frac{1}{V} \int_{\Omega} \mathbf{u}(\mathbf{x}) \cdot \mathbf{v}(\mathbf{x}) dV, \quad \|\mathbf{u}\|_V^2 = \langle \mathbf{u}, \mathbf{u} \rangle_V. \quad (33)$$

The continuous reference domain Ω is discretized into an unstructured tetrahedral mesh $\mathcal{T}_h = \{K_e\}_{e=1}^{N_e}$ using piecewise-linear (P_1) Lagrange continuous finite elements. Within any tetrahe-

dron K , a P_1 displacement field $\mathbf{u}_h(\mathbf{x})$ is an affine vector function of position ($\mathbf{u}_h|_K \in \mathcal{P}_1(K)$). When computing projections against quadratic deformation modes $\mathbf{p}_2 \in \mathcal{P}_2(K)$, the modal cross-product integrand $\mathbf{u}_h \cdot \mathbf{p}_2$ is a cubic polynomial ($\mathcal{P}_3(K)$), and the dictionary Gram matrix integrand $\mathbf{p}_2 \cdot \mathbf{p}_2$ is a quartic polynomial ($\mathcal{P}_4(K)$). Standard 4-point linear tetrahedral quadrature rules (precision degree 2) incur quadrature truncation errors when integrating degree-3 and degree-4 polynomials.

To evaluate all continuum inner products with exact polynomial precision and zero quadrature truncation error, we employ a symmetric, positive degree-4 tetrahedral quadrature rule with 14 points per tetrahedron (constructed via Basix). For each tetrahedron K , the 14 quadrature points $\mathbf{x}_{K,q}$ and strictly positive weights $w_{K,q} > 0$ satisfy $\sum_{q=1}^{14} w_{K,q} = |K|$ and integrate all polynomials up to degree 4 exactly:

$$\int_K p(\mathbf{x}) dV = \sum_{q=1}^{14} w_{K,q} p(\mathbf{x}_{K,q}) \quad \forall p \in \mathcal{P}_4(K). \quad (34)$$

Summing across all N_e tetrahedra gives $N_q = 14N_e$ discrete sampling points with weights $w_p > 0$. Defining normalized quadrature weights $\tilde{w}_p = w_p / \sum_{j=1}^{N_q} w_j$, the continuum inner product (33) is evaluated with exact numerical precision:

$$\langle \mathbf{u}, \mathbf{v} \rangle_V = \sum_{p=1}^{N_q} \tilde{w}_p \mathbf{u}(\mathbf{x}_p) \cdot \mathbf{v}(\mathbf{x}_p). \quad (35)$$

E.3 Infinitesimal rigid-body motion elimination

Infinitesimal rigid-body displacements in $\mathbb{R}^3$ constitute a 6-dimensional linear Lie algebra subspace $\mathfrak{se}(3)$ consisting of 3 translations and 3 rotations:

$$\mathbf{u}_{\text{rigid}}(\mathbf{x}) = \mathbf{t} + \boldsymbol{\omega} \times (\mathbf{x} - \mathbf{x}_c), \quad \mathbf{t}, \boldsymbol{\omega} \in \mathbb{R}^3, \quad (36)$$

where $\mathbf{x}_c$ is the volume-weighted spatial centroid of the pelvic domain,

$$\mathbf{x}_c = \langle \mathbf{x} \rangle_V = \sum_{p=1}^{N_q} \tilde{w}_p \mathbf{x}_p. \quad (37)$$

To ensure scale-free, dimensionally consistent kinematics across subjects, we define the reference gyration radius r_g and non-dimensionalised anatomical coordinates $\tilde{\mathbf{x}} = (\tilde{x}, \tilde{y}, \tilde{z})^\top$:

$$r_g = \sqrt{\langle \|\mathbf{x} - \mathbf{x}_c\|^2 \rangle_V} = \sqrt{\sum_{p=1}^{N_q} \tilde{w}_p \|\mathbf{x}_p - \mathbf{x}_c\|^2}, \quad \tilde{\mathbf{x}} = \frac{\mathbf{x} - \mathbf{x}_c}{r_g}. \quad (38)$$

Here the Cartesian coordinate directions correspond to the standard anatomical axes: x is mediolateral (ML), y is anteroposterior (AP), and z is craniocaudal (CC).

In the discrete quadrature representation, each displacement field is a stacked vector $\mathbf{U} \in \mathbb{R}^{3N_q}$ with components $U_{3p-3+i} = \sqrt{\tilde{w}_p} u_i(\mathbf{x}_p)$ for $i \in \{1, 2, 3\}$. The 6 rigid-body design modes $\mathbf{r}_1, \dots, \mathbf{r}_6 \in \mathbb{R}^{3N_q}$ (corresponding to unit translations along x, y, z and unit infinitesimal rotations about x, y, z) form the rigid design matrix $\mathbf{R} \in \mathbb{R}^{3N_q \times 6}$. We compute its thin QR decomposition,

$$\mathbf{R} = \mathbf{Q}_{\text{rigid}} \mathbf{R}_{\text{upper}}, \quad \mathbf{Q}_{\text{rigid}} \in \mathbb{R}^{3N_q \times 6}, \quad \mathbf{Q}_{\text{rigid}}^\top \mathbf{Q}_{\text{rigid}} = \mathbf{I}_6. \quad (39)$$

The orthogonal projection operator onto the nonrigid displacement subspace is then:

$$\mathbf{P}_{\text{nonrigid}} = \mathbf{I}_{3N_q} - \mathbf{Q}_{\text{rigid}} \mathbf{Q}_{\text{rigid}}^\top. \quad (40)$$

For any raw modal displacement field $\mathbf{u}$, the nonrigid component is $\mathbf{u}_{\text{nonrigid}} = \mathbf{P}_{\text{nonrigid}}\mathbf{u}$, and the total squared displacement norm partitions orthogonally:

$$\|\mathbf{u}\|_V^2 = \|\mathbf{u}_{\text{rigid}}\|_V^2 + \|\mathbf{u}_{\text{nonrigid}}\|_V^2, \quad f_{\text{nonrigid}} = \frac{\|\mathbf{u}_{\text{nonrigid}}\|_V^2}{\|\mathbf{u}\|_V^2}. \quad (41)$$

Crucially, rigid rotation is eliminated at this foundational stage: it is neither retained in the target deformation field nor reported as an explained deformation category. If a displacement field satisfies $\|\mathbf{u}_{\text{nonrigid}}\|_V^2 / \|\mathbf{u}\|_V^2 \leq 10^{-20}$, it is identified as a pure rigid motion and excluded from nonrigid decomposition.

E.4 Canonical 3D polynomial dictionary

The vector space of polynomial displacement fields $\mathbf{u}(\tilde{\mathbf{x}})$ of degree at most 2 in three dimensions has complete algebraic dimension $3 \times \binom{3+2}{2} = 30$. Subtracting the 6 degrees of freedom of infinitesimal rigid motion leaves an exact 24-dimensional nonrigid polynomial quotient space. We decompose this space into four canonical physical deformation families spanning all 24 nonrigid degrees of freedom:

- (i) **Axial family** ($d_{\text{axial}} = 6$ **degrees of freedom**): Uniform normal extension/compression along each anatomical axis, together with quadratic normal strain gradients:

$$\begin{aligned} \mathbf{a}_{x,1} &= (\tilde{x}, 0, 0)^\top, & \mathbf{a}_{x,2} &= (\tfrac{1}{2}\tilde{x}^2, 0, 0)^\top, \\ \mathbf{a}_{y,1} &= (0, \tilde{y}, 0)^\top, & \mathbf{a}_{y,2} &= (0, \tfrac{1}{2}\tilde{y}^2, 0)^\top, \\ \mathbf{a}_{z,1} &= (0, 0, \tilde{z})^\top, & \mathbf{a}_{z,2} &= (0, 0, \tfrac{1}{2}\tilde{z}^2)^\top. \end{aligned} \quad (42)$$

- (ii) **Bending family** ($d_{\text{bending}} = 6$ **degrees of freedom**): Classical pure Euler-Bernoulli flexural modes. For any pair of distinct coordinate axes $i \neq j \in \{x, y, z\}$, the displacement field

$$\mathbf{b}_{ij} = \tfrac{1}{2}\tilde{x}_i^2 \mathbf{e}_j - \tilde{x}_i \tilde{x}_j \mathbf{e}_i \quad (43)$$

satisfies $\partial_i u_j = \tilde{x}_i$ and $\partial_j u_i = -\tilde{x}_i$. Consequently, the Cauchy shear strain vanishes identically:

$$\varepsilon_{ij}(\mathbf{b}_{ij}) = \tfrac{1}{2}(\partial_j u_i + \partial_i u_j) = \tfrac{1}{2}(-\tilde{x}_i + \tilde{x}_i) \equiv 0. \quad (44)$$

The six ordered pairs (x, y) , (x, z) , (y, x) , (y, z) , (z, x) , (z, y) thus define six independent shear-free flexural modes.

- (iii) **Shear family** ($d_{\text{shear}} = 10$ **degrees of freedom**): Comprises three uniform transverse shears, six planar quadratic transverse shear gradients, and one triaxial cross-gradient shear field $\mathbf{w}$:

$$\begin{aligned} \mathbf{s}_{xy,1} &= (\tilde{y}, \tilde{x}, 0)^\top, & \mathbf{s}_{xy,2} &= (\tilde{x}\tilde{y}, \tfrac{1}{2}\tilde{x}^2, 0)^\top, & \mathbf{s}_{yx,2} &= (\tfrac{1}{2}\tilde{y}^2, \tilde{y}\tilde{x}, 0)^\top, \\ \mathbf{s}_{yz,1} &= (0, \tilde{z}, \tilde{y})^\top, & \mathbf{s}_{yz,2} &= (0, \tilde{y}\tilde{z}, \tfrac{1}{2}\tilde{y}^2)^\top, & \mathbf{s}_{zy,2} &= (0, \tfrac{1}{2}\tilde{z}^2, \tilde{z}\tilde{y})^\top, \\ \mathbf{s}_{zx,1} &= (\tilde{z}, 0, \tilde{x})^\top, & \mathbf{s}_{zx,2} &= (\tfrac{1}{2}\tilde{z}^2, 0, \tilde{z}\tilde{x})^\top, & \mathbf{s}_{xz,2} &= (\tilde{x}\tilde{z}, 0, \tfrac{1}{2}\tilde{x}^2)^\top, \\ \mathbf{w} &= (\tilde{y}\tilde{z}, \tilde{x}\tilde{z}, \tilde{x}\tilde{y})^\top = \nabla(\tilde{x}\tilde{y}\tilde{z}). \end{aligned} \quad (45)$$

Unlike bending, these modes produce non-zero shear strains $\varepsilon_{ij} \neq 0$. The triaxial mode $\mathbf{w}$ is irrotational ($\nabla \times \mathbf{w} = \mathbf{0}$) and divergence-free ($\nabla \cdot \mathbf{w} = 0$), uniquely completing the 24-dimensional nonrigid polynomial quotient.

- (iv) **Torsion family** ($d_{\text{torsion}} = 2$ **degrees of freedom**): Saint-Venant torsional twist gradients around the three Cartesian axes:

$$\mathbf{t}_x = (0, -\tilde{x}\tilde{z}, \tilde{x}\tilde{y})^\top, \quad \mathbf{t}_y = (\tilde{y}\tilde{z}, 0, -\tilde{x}\tilde{y})^\top, \quad \mathbf{t}_z = (-\tilde{y}\tilde{z}, \tilde{x}\tilde{z}, 0)^\top. \quad (46)$$

Each torsion field exhibits identically zero normal strain ($\varepsilon_{xx} = \varepsilon_{yy} = \varepsilon_{zz} \equiv 0$). Crucially, these three vector fields satisfy the identity:

$$\mathbf{t}_x + \mathbf{t}_y + \mathbf{t}_z \equiv \mathbf{0}. \quad (47)$$

Hence, the space of Saint-Venant torsional twist has algebraic dimension exactly equal to 2, spanned by $\{\mathbf{t}_x, \mathbf{t}_y\}$.

To assemble the discrete kinematic design matrix, the basis fields in each family $m \in \{\text{axial, bending, shear, torsion}\}$ are sampled at the N_q quadrature points, weighted by $\sqrt{w_p}$, projected orthogonal to $\mathbf{Q}_{\text{rigid}}$ via $\mathbf{P}_{\text{nonrigid}}$, and orthonormalised via reduced singular value decomposition (SVD):

$$\mathbf{P}_{\text{nonrigid}} \mathbf{A}_m = \mathbf{Q}_m \Sigma_m \mathbf{V}_m^\top, \quad \mathbf{Q}_m \in \mathbb{R}^{3N_q \times d_m}, \quad \mathbf{Q}_m^\top \mathbf{Q}_m = \mathbf{I}_{d_m}. \quad (48)$$

The concatenated design matrix $\mathbf{A} = [\mathbf{Q}_{\text{axial}}, \mathbf{Q}_{\text{bending}}, \mathbf{Q}_{\text{shear}}, \mathbf{Q}_{\text{torsion}}] \in \mathbb{R}^{3N_q \times 24}$ spans the complete 24-dimensional canonical nonrigid deformation subspace.

E.5 Numerical conditioning, family non-orthogonality, and the Gram matrix

It is important to note that the four canonical deformation families exhibit intrinsic non-orthogonality and are not mutually orthogonal even on an idealized symmetric domain. For example, on the symmetric unit cube $[-1, 1]^3$ (volume $V = 8$), the unnormalized continuum volume integral between the bending mode $\mathbf{b} = (-xy, \frac{1}{2}x^2, 0)^\top$ and the shear mode $\mathbf{s} = (xy, \frac{1}{2}x^2, 0)^\top$ evaluates analytically to $\int_{[-1, 1]^3} \mathbf{b} \cdot \mathbf{s} dV = -22/45 \approx -0.4889$, corresponding to a volume-normalized inner product $\langle \mathbf{b}, \mathbf{s} \rangle_V = \frac{1}{V} \int_{[-1, 1]^3} \mathbf{b} \cdot \mathbf{s} dV = -11/180 \approx -0.0611$ (angular cosine $\cos \theta = -11/29 \approx -0.3793$). Consequently, the four deformation families share correlated explanatory subspaces by mathematical necessity, independently of geometric irregularity.

On the real, geometrically asymmetric pelvic ring, this intrinsic non-orthogonality combines with anatomical geometry. The kinematic design matrix $\mathbf{A}$ has 2-norm condition number $\kappa(\mathbf{A}) \approx 4.58$, and the associated Gram matrix $\mathbf{G} = \mathbf{A}^\top \mathbf{A} \in \mathbb{R}^{24 \times 24}$ has condition number:

$$\kappa(\mathbf{G}) = \frac{\sigma_{\max}(\mathbf{G})}{\sigma_{\min}(\mathbf{G})} = \kappa(\mathbf{A})^2 \approx 20.98. \quad (49)$$

A Gram condition number of ~ 21 confirms that all 24 canonical modes remain linearly independent and well-conditioned without numerical rank deficiency.

E.6 Cooperative game-theoretic Shapley value norm allocation

Because the four deformation subspaces exhibit intrinsic non-orthogonality (shared explained norm) and cannot be mutually orthogonal, projecting onto them sequentially (e.g., Gram-Schmidt orthogonalisation) would introduce an arbitrary, order-dependent bias: the first chosen family would absorb all shared variance, artificially starving subsequent families. To achieve an objective, order-independent decomposition, we formulate the problem using cooperative game theory.

Let $M = \{\text{axial, bending, shear, torsion}\}$ denote the set of four pattern players ($|M| = 4$). There are $2^4 = 16$ possible coalitions $S \subseteq M$. For any coalition S , we construct the concatenated family design matrix $\mathbf{A}_S \in \mathbb{R}^{3N_q \times \sum_{m \in S} d_m}$ and define the characteristic function $v(S)$ as the fraction of nonrigid squared displacement norm explained by the joint subspace:

$$v(S) = R^2(S) = \frac{\mathbf{u}_{\text{nonrigid}}^\top \mathbf{P}_S \mathbf{u}_{\text{nonrigid}}}{\|\mathbf{u}_{\text{nonrigid}}\|_V^2}, \quad \mathbf{P}_S = \mathbf{A}_S (\mathbf{A}_S^\top \mathbf{A}_S)^+ \mathbf{A}_S^\top, \quad (50)$$

with $v(\emptyset) = 0$, where $(\cdot)^+$ denotes the Moore-Penrose pseudoinverse. The Shapley value f_m assigned to each deformation family $m \in M$ is the unique allocation satisfying the four foundational Shapley axioms:

1. **Efficiency:** The sum of allocated values equals the value of the grand coalition:

$$\sum_{m \in M} f_m = v(M) = R^2(M) = 1 - f_{\text{residual}}. \quad (51)$$

2. **Symmetry:** If two families $m, k \in M$ contribute identically to every coalition ($v(S \cup \{m\}) = v(S \cup \{k\})$ for all $S \subseteq M \setminus \{m, k\}$), then $f_m = f_k$.
3. **Dummy player:** If a family m adds zero marginal variance to every coalition ($v(S \cup \{m\}) = v(S)$ for all $S \subseteq M \setminus \{m\}$), then $f_m = 0$.
4. **Additivity:** For any two characteristic value functions v and w defined on the same set of players, the values allocated under the combined sum game $(v + w)(S) = v(S) + w(S)$ satisfy $\phi_m(v + w) = \phi_m(v) + \phi_m(w)$.

The explicit Shapley allocation formula evaluates the marginal contribution of family m averaged over all coalition orders:

$$f_m = \sum_{S \subseteq M \setminus \{m\}} \frac{|S|!(|M| - |S| - 1)!}{|M|!} [v(S \cup \{m\}) - v(S)]. \quad (52)$$

The unexplained shape change is retained as an explicit residual:

$$f_{\text{residual}} = 1 - v(M) = 1 - R^2(\text{axial} \cup \text{bending} \cup \text{shear} \cup \text{torsion}). \quad (53)$$

Because subspace projection operators are monotonic ($V_S \subseteq V_{S \cup \{m\}} \implies v(S \cup \{m\}) \geq v(S)$), every marginal contribution is non-negative, which guarantees $f_m \geq 0$ for all m . Exact closure holds identically:

$$f_{\text{axial}} + f_{\text{bending}} + f_{\text{shear}} + f_{\text{torsion}} + f_{\text{residual}} \equiv 1.0000 \quad (100.0\%). \quad (54)$$

Furthermore, the fractions are strictly scale-invariant ($\mathbf{u} \mapsto c\mathbf{u}$ for $c \neq 0$) and sign-invariant ($\mathbf{u} \mapsto -\mathbf{u}$). For comparative diagnostics, we also compute the standalone coefficient of determination for each isolated family:

$$R_m^2 = v(\{m\}) = \frac{\mathbf{u}_{\text{nonrigid}}^\top \mathbf{P}_m \mathbf{u}_{\text{nonrigid}}}{\|\mathbf{u}_{\text{nonrigid}}\|_V^2}. \quad (55)$$

Due to the intrinsic geometric coupling between families ($\kappa(\mathbf{G}) \approx 20.98$), the sum $\sum_{m \in M} R_m^2$ may moderately exceed the total explained fraction $R^2(M)$; the Shapley values partition this shared overlap fairly and symmetrically.

E.7 Synthetic benchmark and backwards-identification validation

To verify the accuracy, robustness, and backwards-identifiability of the global deformation decomposition, we executed synthetic validation experiments on both an idealized symmetric domain (cube) and the realistic reference pelvis mesh (Table E2).

On the symmetric domain, pure synthetic fields belonging to the axial, shear, and torsion families are identified with 100.0% fidelity, zero cross-talk, and exactly 0.0% residual. For pure Euler-Bernoulli bending ($\mathbf{u} = (-xy, \frac{1}{2}x^2, 0)$), the displacement field consists of an axial cross-term $-xy$ and a transverse parabolic deflection $\frac{1}{2}x^2$. Because the quadratic shear mode

$\mathbf{s}_{xy,2} = (xy, \frac{1}{2}x^2, 0)$ shares the identical transverse parabolic deflection $\frac{1}{2}x^2$, the pure flexural field lies in the joint span $V_{\text{bending}} \oplus V_{\text{shear}}$. The Shapley allocation splits this shared transverse deflection analytically, yielding 77.5% bending, 22.5% shear, and exactly 0.0% residual (100.0% fitted).

On the actual reference pelvis mesh, pure canonical polynomial fields are recovered with exactly 0.0% residual (100.0% fitted fraction):

- Pure axial stretch ($x\mathbf{e}_x$): 85.3% axial share, 0.0% residual (100.0% fitted).
- Pure triaxial extension ($x\mathbf{e}_x - \frac{1}{2}y\mathbf{e}_y - \frac{1}{2}z\mathbf{e}_z$): 84.8% axial share, 0.0% residual (100.0% fitted).
- Pure Euler-Bernoulli bending: 74.1% bending share, 0.0% residual (100.0% fitted).
- Pure transverse shear: 82.8–90.7% shear share, 0.0% residual (100.0% fitted).
- Pure Saint-Venant torsion: 91.3–97.7% torsion share, 0.0% residual (100.0% fitted).

When subjected to pure infinitesimal rigid translations or pure infinitesimal rotations, the non-rigid fraction is detected as $f_{\text{nonrigid}} < 10^{-27}$, correctly triggering the rigid-motion filter and confirming zero spurious contamination. Finally, adding arbitrary rigid-body translations and rotations to an axial deformation field yields identical Shapley shares (85.3% axial, 7.2% bending, 7.5% shear, 0.0% torsion, 0.0% residual) to within machine precision ($< 10^{-10}$), demonstrating complete invariance under rigid-body frame shifts.

E.8 Reference mode deformation profiles and biomechanical interpretation

Table E1 compiles the quantitative whole-pelvis 3D deformation decomposition for all fifteen reference eigenmodes. The results resolve several key biomechanical features of the pelvic ring:

- **Modes 1 and 2 (low-rank backbone):** For Mode 1, the nonrigid deformation accounts for 3.3% of the total displacement norm (the remainder reflecting whole-pelvis rocking on the sacroiliac/pubis supports). Its nonrigid shape change is predominantly out-of-plane torsional twist (22.7%) and transverse shear (17.7%), with a total fitted fraction of 42.7% (residual 57.3%). For Mode 2, nonrigid shape change (5.5% of total norm) is a coupled sagittal ring flexure and shear (23.6% bending, 23.7% shear, 20.8% axial, 68.6% fitted, residual 31.4%). This resolves the earlier 1D beam artifact (which had yielded an unphysical 84% residual): the true 3D ring kinematics explain more than two-thirds of the deformation norm.
- **Modes 4 and 5 (dominant flexural pathways):** Modes 4 and 5 exhibit strong bending dominance, explaining 43.6% and 47.4% of their nonrigid norm respectively (with standalone bending fits R_{bending}^2 reaching 47.6% and 62.4%, and total explained fractions of 74.4% and 86.3%). These modes capture lateral and anteroposterior flaring of the iliac wings.
- **The obstetric 9–10 pair:** Mode 9 is the most highly fitted mode in the entire spectrum (91.8% explained, residual only 8.2%). Its deformation is overwhelmingly axial (60.6% Shapley share; standalone $R_{\text{axial}}^2 = 67.7\%$), representing circumferential normal stretching that widens the obstetric pelvic inlet. In contrast, its near-degenerate partner Mode 10 (74.9% fitted) is dominated by shear (35.2%) and torsion (34.3%), with minimal axial contribution (4.3%; standalone $R_{\text{torsion}}^2 = 30.3\%$, $R_{\text{shear}}^2 = 32.7\%$, residual 25.1%). This quantitative kinematic divergence reinforces why the 9–10 cluster forms a complementary functional unit: Mode 9 provides direct obstetric diameter enlargement, whereas Mode 10 accommodates asymmetric torsional/shear loads.

Table E1. Reference mode whole-pelvis 3D deformation decomposition

Mode	Nonrigid	Fitted	Shapley norm partition (%)					Standalone R^2 (%)			
			Axial	Bending	Shear	Torsion	Residual	Axial	Bending	Shear	Torsion
1	3.3	42.7	0.9	1.5	17.7	22.7	57.3	0.4	2.2	15.4	19.0
2	5.5	68.6	20.8	23.6	23.7	0.5	31.4	15.4	16.1	16.9	0.8
3	29.0	76.7	2.8	4.7	41.7	27.5	23.3	5.1	5.1	52.7	37.4
4	24.5	74.4	13.5	43.6	15.8	1.5	25.6	24.6	47.6	14.9	2.2
5	73.0	86.3	4.3	47.4	24.2	10.4	13.7	5.4	62.4	38.7	13.7
6	71.9	82.3	3.1	26.9	24.9	27.3	17.7	2.2	33.0	37.2	35.2
7	69.5	72.1	16.9	7.3	44.3	3.5	27.9	29.4	4.6	55.8	4.1
8	57.0	72.1	22.1	20.3	29.5	0.2	27.9	32.7	32.5	41.3	0.4
9	92.1	91.8	60.6	18.4	12.2	0.5	8.2	67.7	31.6	14.9	0.3
10	95.7	74.9	4.3	1.1	35.2	34.3	25.1	6.2	1.5	32.7	30.3
11	71.4	43.1	5.4	3.9	21.1	12.8	56.9	2.8	0.4	9.5	12.2
12	94.2	62.3	9.6	22.4	27.4	3.0	37.7	13.1	12.7	20.3	2.4
13	98.3	48.4	9.2	19.9	19.1	0.2	51.6	8.3	27.0	25.7	0.0
14	92.0	53.4	1.2	15.9	33.4	3.0	46.6	0.9	5.6	24.9	4.7
15	98.7	55.6	18.4	23.5	13.7	0.0	44.4	16.1	28.2	13.7	0.0

Table E2. Synthetic benchmark and backwards-identification validation

Domain	Synthetic deformation field	Axial (%)	Bending (%)	Shear (%)	Torsion (%)	Resid. (%)	Outcome
Symmetric cube	Pure axial stretch ($x\mathbf{e}_x$)	100.0	0.0	0.0	0.0	0.0	Exact closure (0.0% residual)
Symmetric cube	Pure bending ($-xy\mathbf{e}_x + \frac{1}{2}x^2\mathbf{e}_y$)	0.0	77.5	22.5	0.0	0.0	Exact closure (0.0% residual)
Symmetric cube	Pure shear ($y\mathbf{e}_x + x\mathbf{e}_y$)	0.0	0.0	100.0	0.0	0.0	Exact closure (0.0% residual)
Symmetric cube	Pure torsion ($yz\mathbf{e}_x - xy\mathbf{e}_z$)	0.0	0.0	0.0	100.0	0.0	Exact closure (0.0% residual)
Pelvis mesh	Pure axial stretch ($x\mathbf{e}_x$)	85.3	7.2	7.5	0.0	0.0	Exact closure (0.0% residual)
Pelvis mesh	Pure triaxial ($x\mathbf{e}_x - \frac{1}{2}y\mathbf{e}_y - \frac{1}{2}z\mathbf{e}_z$)	84.8	11.1	4.1	0.0	0.0	Exact closure (0.0% residual)
Pelvis mesh	Pure bending ($-xy\mathbf{e}_x + \frac{1}{2}x^2\mathbf{e}_y$)	17.1	74.1	8.8	0.0	0.0	Exact closure (0.0% residual)
Pelvis mesh	Pure shear ($y\mathbf{e}_x + x\mathbf{e}_y$)	13.7	1.8	82.8	1.7	0.0	Exact closure (0.0% residual)
Pelvis mesh	Pure torsion ($yz\mathbf{e}_x - xy\mathbf{e}_z$)	5.5	0.5	7.0	87.1	0.0	Exact closure (0.0% residual)
Pelvis mesh	Infinitesimal rigid translation	0.0	0.0	0.0	0.0	0.0	Rigid: rejected ($f_{\text{nonrigid}} < 10^{-20}$)
Pelvis mesh	Infinitesimal rigid rotation	0.0	0.0	0.0	0.0	0.0	Rigid: rejected ($f_{\text{nonrigid}} < 10^{-20}$)
Pelvis mesh	Axial stretch + rigid perturbation	85.3	7.2	7.5	0.0	0.0	Exact closure (0.0% residual)

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